

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent	12398574
Kind Code	B2
Date of Patent	August 26, 2025
Inventor(s)	Pickering; Samuel Thomas

Adhesive application system

Abstract

An adhesive application system for an automated block laying machine, the adhesive application system including: (a) an adhesive supply; and, (b) an adhesive applicator including: (i) a container for storing an adhesive reservoir received from the adhesive supply via an adhesive supply line; (ii) a lid mounted to the container so as to substantially cover the adhesive reservoir; and, (iii) an adhesive roller rotatably coupled to the container so that upon rotation adhesive from the reservoir is entrained up and around at least a portion of the adhesive roller, wherein, in use, adhesive is transferred from the adhesive roller onto a bottom surface of a block as the block is moved across the roller.

Inventors:	Pickering; Samuel Thomas (Kalamunda, AU)
Applicant:	FASTBRICK IP PTY LTD (High Wycombe, AU)
Family ID:	1000008777795
Assignee:	FASTBRICK IP PTY LTD (High Wycombe, AU)
Appl. No.:	18/004277
Filed (or PCT Filed):	July 08, 2021
PCT No.:	PCT/AU2021/050732
PCT Pub. No.:	WO2022/006635
PCT Pub. Date:	January 13, 2022

Prior Publication Data

Document Identifier	Publication Date
US 20240328180 A1	Oct. 03, 2024

Foreign Application Priority Data

AU

2020902367

Jul. 08, 2020

Publication Classification

Int. Cl.: E04G21/22 (20060101); B25J15/00 (20060101); B25J18/02 (20060101); E04G21/20 (20060101)

U.S. Cl.:

CPC E04G21/22 (20130101); B25J15/0061 (20130101); B25J15/0066 (20130101); B25J15/008 (20130101); B25J18/025 (20130101); E04G21/20 (20130101);

Field of Classification Search

CPC: E04G (21/20); E04G (21/22); B25J (15/0061); B25J (15/0066); B25J (15/008); B25J (18/025); E04F (21/023)

USPC: 52/749.14

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
428096	12/1889	Hathaway	431/328	F23D 14/16
1411172	12/1921	Kaye	414/792.5	E04G 21/22
1633192	12/1926	Reagan	N/A	N/A
1829435	12/1930	Barnhart	N/A	N/A
2072663	12/1936	Bedford	219/421	A21C 15/002
2599552	12/1951	Harney	414/792.5	E04G 21/22
3162886	12/1963	Wise	401/35	E04G 21/204
3438171	12/1968	Demarest	N/A	N/A
3746570	12/1972	McIntosh	427/208.2	B05C 11/10
3757484	12/1972	Williamson et al.	N/A	N/A
3786966	12/1973	Behunin	401/187	E04G 21/202
3790428	12/1973	Lingl	N/A	N/A
3791559	12/1973	Foye	222/240	E04G 21/204
3834973	12/1973	Kummerow	156/182	E04C 2/042
RE28305	12/1974	Williamson et al.	N/A	N/A
3930929	12/1975	Lingl	N/A	N/A
3950914	12/1975	Lowen	N/A	N/A
4033463	12/1976	Cervin	N/A	N/A
4039170	12/1976	Cornwell	366/29	A62C 5/00

4043487	12/1976	Price	425/456	E04G 21/204
4067766	12/1977	Larger	52/749.14	E04C 2/042
4106259	12/1977	Taylor-smith	N/A	N/A
4135651	12/1978	Hession	401/48	E04G 21/204
4221258	12/1979	Richard	N/A	N/A
4245451	12/1980	Taylor-smith	N/A	N/A
4303363	12/1980	Cervin	N/A	N/A
4523100	12/1984	Payne	N/A	N/A
4634347	12/1986	Gauffin	417/900	E04G 21/20
4635985	12/1986	Rooke	N/A	N/A
4708562	12/1986	Melan et al.	N/A	N/A
4714339	12/1986	Lau	N/A	N/A
4758036	12/1987	Legille et al.	N/A	N/A
4765789	12/1987	Lonardi et al.	N/A	N/A
4790651	12/1987	Brown et al.	N/A	N/A
4827689	12/1988	Lonardi et al.	N/A	N/A
4829737	12/1988	Anliker	52/749.14	E04G 21/22
4852237	12/1988	Tradl et al.	N/A	N/A
4911595	12/1989	Kirchen et al.	N/A	N/A
4945493	12/1989	Huang et al.	N/A	N/A
4952772	12/1989	Zana	N/A	N/A
4954762	12/1989	Miyake et al.	N/A	N/A
4969789	12/1989	Searle	N/A	N/A
5004844	12/1990	Van Leeuwen et al.	N/A	N/A
5013986	12/1990	Gauggel	N/A	N/A
5018923	12/1990	Melan et al.	N/A	N/A
5035352	12/1990	Mania	222/614	E04G 21/20
5049797	12/1990	Phillips	N/A	N/A
5080415	12/1991	Bjornson	N/A	N/A
5193723	12/1992	Everett	222/615	E04G 21/20
5196900	12/1992	Pettersen	N/A	N/A
5284000	12/1993	Milne et al.	N/A	N/A
5321353	12/1993	Furness	N/A	N/A
5403140	12/1994	Carmichael et al.	N/A	N/A
5413454	12/1994	Movsesian	N/A	N/A
5419669	12/1994	Kremer et al.	N/A	N/A
5420489	12/1994	Hansen et al.	N/A	N/A
5469531	12/1994	Faure et al.	N/A	N/A
5497061	12/1995	Nonaka et al.	N/A	N/A
5523663	12/1995	Tsuge et al.	N/A	N/A
5527145	12/1995	Duncan	N/A	N/A
5557397	12/1995	Hyde et al.	N/A	N/A
5581975	12/1995	Trebbi	N/A	N/A
5681125	12/1996	Amaya	401/265	E04G 21/20
5737500	12/1997	Seraji et al.	N/A	N/A
5743374	12/1997	Monsees	N/A	N/A
5838882	12/1997	Gan et al.	N/A	N/A

5878921	12/1998	Chase	222/626	E04F 21/08
5895690	12/1998	Greisel	N/A	N/A
6018923	12/1999	Wendt	N/A	N/A
6049377	12/1999	Lau et al.	N/A	N/A
6101455	12/1999	Davis	N/A	N/A
6112955	12/1999	Lang	222/413	E04G 21/20
6134507	12/1999	Markey, Jr. et al.	N/A	N/A
6166809	12/1999	Pettersen et al.	N/A	N/A
6166811	12/1999	Long et al.	N/A	N/A
6170221	12/2000	Behunin	52/745.1	E04G 21/20
6172754	12/2000	Niebuhr	N/A	N/A
6213309	12/2000	Dadisho	N/A	N/A
6285959	12/2000	Greer	N/A	N/A
6310644	12/2000	Keightley	N/A	N/A
6330503	12/2000	Sharp et al.	N/A	N/A
6370837	12/2001	Mcmahon et al.	N/A	N/A
6427122	12/2001	Lin	N/A	N/A
6429016	12/2001	Mcneil	N/A	N/A
6512993	12/2002	Kacyra et al.	N/A	N/A
6516272	12/2002	Lin	N/A	N/A
6584378	12/2002	Anfindsen	N/A	N/A
6611141	12/2002	Schulz	N/A	N/A
6618496	12/2002	Tassakos et al.	N/A	N/A
6628322	12/2002	Cerruti	N/A	N/A
6643002	12/2002	Drake, Jr.	N/A	N/A
6664529	12/2002	Pack et al.	N/A	N/A
6681145	12/2003	Greenwood et al.	N/A	N/A
6683694	12/2003	Cornil	N/A	N/A
6704619	12/2003	Coleman et al.	N/A	N/A
6741364	12/2003	Lange et al.	N/A	N/A
6825937	12/2003	Gebauer et al.	N/A	N/A
6850946	12/2004	Rappaport et al.	N/A	N/A
6859729	12/2004	Breakfield et al.	N/A	N/A
6864966	12/2004	Giger	N/A	N/A
6868847	12/2004	Ainedter et al.	N/A	N/A
6873880	12/2004	Hooke et al.	N/A	N/A
6917893	12/2004	Dietsch et al.	N/A	N/A
6935036	12/2004	Barber et al.	N/A	N/A
6957496	12/2004	Raab et al.	N/A	N/A
6965843	12/2004	Hobden et al.	N/A	N/A
6970802	12/2004	Ban et al.	N/A	N/A
6996912	12/2005	Raab et al.	N/A	N/A
7050930	12/2005	Hobden et al.	N/A	N/A
7051450	12/2005	Barber et al.	N/A	N/A
7069664	12/2005	Barber et al.	N/A	N/A
7107144	12/2005	Capozzi et al.	N/A	N/A
7111437	12/2005	Ainedter	N/A	N/A
7130034	12/2005	Barvosa-carter et al.	N/A	N/A
7142981	12/2005	Hablani	N/A	N/A

7145647	12/2005	Suphellen et al.	N/A	N/A
7153454	12/2005	Khoshnevis	N/A	N/A
7174651	12/2006	Barber et al.	N/A	N/A
7230689	12/2006	Lau	N/A	N/A
7246030	12/2006	Raab et al.	N/A	N/A
7269910	12/2006	Raab et al.	N/A	N/A
7347311	12/2007	Rudge	N/A	N/A
7519493	12/2008	Atwell et al.	N/A	N/A
7551121	12/2008	Oconnell et al.	N/A	N/A
7564538	12/2008	Sakimura et al.	N/A	N/A
7570371	12/2008	Storm	N/A	N/A
7576836	12/2008	Bridges	N/A	N/A
7576847	12/2008	Bridges	N/A	N/A
7591078	12/2008	Crampton	N/A	N/A
7639347	12/2008	Eaton	N/A	N/A
7693325	12/2009	Pulla et al.	N/A	N/A
7701587	12/2009	Shioda et al.	N/A	N/A
7743586	12/2009	Hancock	222/614	E04B 2/16
7774159	12/2009	Cheng et al.	N/A	N/A
7800758	12/2009	Bridges et al.	N/A	N/A
7804602	12/2009	Raab	N/A	N/A
RE42055	12/2010	Raab et al.	N/A	N/A
RE42082	12/2010	Raab et al.	N/A	N/A
7881896	12/2010	Atwell et al.	N/A	N/A
7967549	12/2010	Geist et al.	N/A	N/A
7993289	12/2010	Quistgaard et al.	N/A	N/A
8036452	12/2010	Pettersson et al.	N/A	N/A
8054451	12/2010	Karazi et al.	N/A	N/A
8060344	12/2010	Stathis	N/A	N/A
8145446	12/2011	Atwell et al.	N/A	N/A
8166727	12/2011	Pivac et al.	N/A	N/A
8169604	12/2011	Braghiroli et al.	N/A	N/A
8185240	12/2011	Williams et al.	N/A	N/A
8229208	12/2011	Pulla et al.	N/A	N/A
8233153	12/2011	Knuettel	N/A	N/A
8244030	12/2011	Pettersson et al.	N/A	N/A
8248620	12/2011	Wicks et al.	N/A	N/A
8269984	12/2011	Hinderling et al.	N/A	N/A
8327555	12/2011	Champ	N/A	N/A
8337407	12/2011	Quistgaard et al.	N/A	N/A
8345926	12/2012	Clark et al.	N/A	N/A
8346392	12/2012	Walser et al.	N/A	N/A
8405716	12/2012	Yu et al.	N/A	N/A
8467072	12/2012	Cramer et al.	N/A	N/A
8537372	12/2012	Siercks et al.	N/A	N/A
8537376	12/2012	Day et al.	N/A	N/A
8558992	12/2012	Steffey	N/A	N/A
8593648	12/2012	Cramer et al.	N/A	N/A
8595948	12/2012	Raab et al.	N/A	N/A
8606399	12/2012	Williams et al.	N/A	N/A

8634950	12/2013	Simonetti et al.	N/A	N/A
8644964	12/2013	Hendron et al.	N/A	N/A
8668074	12/2013	Davidson	N/A	N/A
8670114	12/2013	Bridges et al.	N/A	N/A
8677643	12/2013	Bridges et al.	N/A	N/A
8792709	12/2013	Pulla et al.	N/A	N/A
8803055	12/2013	Lau et al.	N/A	N/A
8812155	12/2013	Brethe	N/A	N/A
8825208	12/2013	Benson	700/247	E04F 21/02
8832954	12/2013	Atwell et al.	N/A	N/A
8848203	12/2013	Bridges et al.	N/A	N/A
8875409	12/2013	Kretschmer et al.	N/A	N/A
8898919	12/2013	Bridges et al.	N/A	N/A
8902408	12/2013	Bridges	N/A	N/A
8913814	12/2013	Gandyra	N/A	N/A
8931182	12/2014	Raab et al.	N/A	N/A
8942940	12/2014	York	N/A	N/A
8965571	12/2014	Peters et al.	N/A	N/A
8996244	12/2014	Summer et al.	N/A	N/A
8997362	12/2014	Briggs et al.	N/A	N/A
9020240	12/2014	Pettersson et al.	N/A	N/A
9028908	12/2014	DeTar	427/119	C23C 2/185
9033998	12/2014	Schaible et al.	N/A	N/A
RE45565	12/2014	Bridges et al.	N/A	N/A
9046360	12/2014	Atwell et al.	N/A	N/A
9074381	12/2014	Drew	N/A	N/A
9109877	12/2014	Thierman	N/A	N/A
9146315	12/2014	Bosse et al.	N/A	N/A
9151830	12/2014	Bridges	N/A	N/A
9157246	12/2014	Campbell	N/A	E04G 21/202
9163922	12/2014	Bridges et al.	N/A	N/A
9170096	12/2014	Fowler et al.	N/A	N/A
9188430	12/2014	Atwell et al.	N/A	N/A
9207309	12/2014	Bridges	N/A	N/A
9223025	12/2014	Debrunner et al.	N/A	N/A
9229108	12/2015	Debrunner et al.	N/A	N/A
9266238	12/2015	Huettenhofer	N/A	N/A
9267784	12/2015	Atwell et al.	N/A	N/A
9278448	12/2015	Freeman	N/A	N/A
9279661	12/2015	Tateno et al.	N/A	N/A
9303988	12/2015	Tani	N/A	N/A
9353519	12/2015	Williams	N/A	N/A
9354051	12/2015	Dunne et al.	N/A	N/A
9358688	12/2015	Drew	N/A	N/A
9367741	12/2015	Le Marec	N/A	N/A
9377301	12/2015	Neier et al.	N/A	N/A
9383200	12/2015	Hulm et al.	N/A	N/A
9395174	12/2015	Bridges	N/A	N/A
9405293	12/2015	Meuleau	N/A	N/A

9423282	12/2015	Moy	N/A	N/A
9437005	12/2015	Tateno et al.	N/A	N/A
9443308	12/2015	Pettersson et al.	N/A	N/A
9452533	12/2015	Calkins et al.	N/A	N/A
9454818	12/2015	Cramer	N/A	N/A
9476695	12/2015	Becker et al.	N/A	N/A
9481007	12/2015	Rzonca	N/A	B05C 11/1002
9482524	12/2015	Metzler et al.	N/A	N/A
9482525	12/2015	Bridges	N/A	N/A
9482746	12/2015	Bridges	N/A	N/A
9494686	12/2015	Maryfield et al.	N/A	N/A
9513100	12/2015	Raab et al.	N/A	N/A
9536163	12/2016	Veeser et al.	N/A	N/A
9541371	12/2016	Pettersson et al.	N/A	N/A
9561019	12/2016	Mihailescu et al.	N/A	N/A
9607239	12/2016	Bridges et al.	N/A	N/A
9618620	12/2016	Zweigle et al.	N/A	N/A
9658061	12/2016	Wilson et al.	N/A	N/A
9671221	12/2016	Ruhland et al.	N/A	N/A
9679385	12/2016	Suzuki et al.	N/A	N/A
9686532	12/2016	Tohme	N/A	N/A
9708079	12/2016	Desjardien et al.	N/A	N/A
9715730	12/2016	Suzuki	N/A	N/A
9720087	12/2016	Christen et al.	N/A	N/A
9734609	12/2016	Pulla et al.	N/A	N/A
9739595	12/2016	Lau	N/A	N/A
9746308	12/2016	Gong	N/A	N/A
9757859	12/2016	Kolb et al.	N/A	N/A
9768837	12/2016	Charvat et al.	N/A	N/A
9772173	12/2016	Atwell et al.	N/A	N/A
9803969	12/2016	Gong	N/A	N/A
9816813	12/2016	Lettau et al.	N/A	N/A
9829305	12/2016	Gong	N/A	N/A
9835717	12/2016	Bosse et al.	N/A	N/A
9844792	12/2016	Pettersson et al.	N/A	N/A
9879976	12/2017	Bridges et al.	N/A	N/A
9897442	12/2017	Pettersson et al.	N/A	N/A
9903939	12/2017	Charvat et al.	N/A	N/A
9909855	12/2017	Becker et al.	N/A	N/A
9915733	12/2017	Fried et al.	N/A	N/A
9921046	12/2017	Gong	N/A	N/A
9958268	12/2017	Ohtomo et al.	N/A	N/A
9958545	12/2017	Eichenholz et al.	N/A	N/A
9964398	12/2017	Becker et al.	N/A	N/A
9964402	12/2017	Tohme et al.	N/A	N/A
9967545	12/2017	Tohme	N/A	N/A
9989353	12/2017	Bartmann et al.	N/A	N/A
10012732	12/2017	Eichenholz et al.	N/A	N/A
10030972	12/2017	Iseli et al.	N/A	N/A

10041793	12/2017	Metzler et al.	N/A	N/A
10054422	12/2017	Böckem et al.	N/A	N/A
10058394	12/2017	Johnson et al.	N/A	N/A
10073162	12/2017	Charvat et al.	N/A	N/A
10074889	12/2017	Charvat et al.	N/A	N/A
10082521	12/2017	Atlas et al.	N/A	N/A
10090944	12/2017	Charvat et al.	N/A	N/A
10094909	12/2017	Charvat et al.	N/A	N/A
10126415	12/2017	Becker et al.	N/A	N/A
10189176	12/2018	Williams	N/A	N/A
10220511	12/2018	Linnell et al.	N/A	N/A
10240949	12/2018	Peters et al.	N/A	N/A
10315904	12/2018	Landler	N/A	N/A
10635758	12/2019	Pivac	N/A	B25J 19/022
10865578	12/2019	Pivac et al.	N/A	N/A
10876308	12/2019	Pivac et al.	N/A	N/A
11106836	12/2020	Pivac	N/A	B28D 7/005
11401115	12/2021	Pivac et al.	N/A	N/A
2002/0030145	12/2001	Lang	248/131	E04G 21/025
2002/0176603	12/2001	Bauer et al.	N/A	N/A
2003/0048459	12/2002	Gooch	N/A	N/A
2003/0090682	12/2002	Gooch et al.	N/A	N/A
2003/0120377	12/2002	Hooke et al.	N/A	N/A
2003/0206285	12/2002	Lau	N/A	N/A
2004/0078137	12/2003	Breakfield et al.	N/A	N/A
2004/0093119	12/2003	Gunnarsson et al.	N/A	N/A
2004/0200947	12/2003	Lau	N/A	N/A
2005/0007450	12/2004	Hill et al.	N/A	N/A
2005/0057745	12/2004	Bontje	N/A	N/A
2005/0060092	12/2004	Hablani	N/A	N/A
2005/0086901	12/2004	Chisholm	N/A	N/A
2005/0131619	12/2004	Rappaport et al.	N/A	N/A
2005/0196484	12/2004	Khoshnevis	N/A	N/A
2005/0252118	12/2004	Matsufuji	N/A	N/A
2006/0167587	12/2005	Read	N/A	N/A
2006/0215179	12/2005	Mcmurtry et al.	N/A	N/A
2007/0024870	12/2006	Girard et al.	N/A	N/A
2007/0229802	12/2006	Lau	N/A	N/A
2007/0284215	12/2006	Rudge	N/A	N/A
2008/0030855	12/2007	Lau	N/A	N/A
2008/0189046	12/2007	Eliasson et al.	N/A	N/A
2009/0038258	12/2008	Pivac et al.	N/A	N/A
2009/0074979	12/2008	Krogedal et al.	N/A	N/A
2010/0025349	12/2009	Khoshnevis	N/A	N/A
2010/0138185	12/2009	Kang	N/A	N/A
2010/0274390	12/2009	Walser et al.	N/A	N/A
2010/0281822	12/2009	Murray	N/A	N/A
2011/0066393	12/2010	Groll et al.	N/A	N/A

2011/0153524	12/2010	Schnackel	N/A	N/A
2011/0208347	12/2010	Otake et al.	N/A	N/A
2012/0038074	12/2011	Khoshnevis	N/A	N/A
2012/0099096	12/2011	Bridges et al.	N/A	N/A
2012/0136524	12/2011	Everett et al.	N/A	N/A
2012/0265391	12/2011	Letsky	N/A	N/A
2012/0277898	12/2011	Kawai et al.	N/A	N/A
2013/0028478	12/2012	St-pierre et al.	N/A	N/A
2013/0068061	12/2012	Yoon	N/A	N/A
2013/0103192	12/2012	Huettenhofer	N/A	N/A
2013/0104407	12/2012	Lee	N/A	N/A
2013/0222816	12/2012	Briggs et al.	N/A	N/A
2013/0250285	12/2012	Bridges et al.	N/A	N/A
2013/0286196	12/2012	Atwell	N/A	N/A
2014/0002608	12/2013	Atwell et al.	N/A	N/A
2014/0067121	12/2013	Brooks et al.	N/A	N/A
2014/0076923	12/2013	Clark	73/304C	B05C 11/1042
2014/0176677	12/2013	Valkenburg et al.	N/A	N/A
2014/0192187	12/2013	Atwell et al.	N/A	N/A
2014/0309960	12/2013	Vennegeerts et al.	N/A	N/A
2014/0343727	12/2013	Calkins et al.	N/A	N/A
2014/0348388	12/2013	Metzler et al.	N/A	N/A
2014/0366481	12/2013	Benson	N/A	N/A
2015/0082740	12/2014	Peters et al.	N/A	N/A
2015/0100066	12/2014	Kostrzewski et al.	N/A	N/A
2015/0134303	12/2014	Chang et al.	N/A	N/A
2015/0153720	12/2014	Pettersson et al.	N/A	N/A
2015/0158181	12/2014	Kawamura et al.	N/A	N/A
2015/0241203	12/2014	Jordil et al.	N/A	N/A
2015/0258694	12/2014	Hand et al.	N/A	N/A
2015/0276402	12/2014	Grsser et al.	N/A	N/A
2015/0280829	12/2014	Breuer	N/A	N/A
2015/0293596	12/2014	Krausen et al.	N/A	N/A
2015/0309175	12/2014	Hinderling et al.	N/A	N/A
2015/0314890	12/2014	Desjardien et al.	N/A	N/A
2015/0352721	12/2014	Wicks et al.	N/A	N/A
2015/0355310	12/2014	Gong et al.	N/A	N/A
2015/0367509	12/2014	Georgeson	N/A	N/A
2015/0371082	12/2014	Csaszar et al.	N/A	N/A
2015/0377606	12/2014	Thielemans	N/A	N/A
2016/0005185	12/2015	Geissler	N/A	N/A
2016/0093099	12/2015	Bridges	N/A	N/A
2016/0153786	12/2015	Liu et al.	N/A	N/A
2016/0187130	12/2015	Metzler et al.	N/A	N/A
2016/0187470	12/2015	Becker et al.	N/A	N/A
2016/0223364	12/2015	Peters et al.	N/A	N/A
2016/0242744	12/2015	Mihailescu et al.	N/A	N/A

2016/0263767	12/2015	Williams	N/A	N/A
2016/0274237	12/2015	Stutz	N/A	N/A
2016/0282107	12/2015	Roland et al.	N/A	N/A
2016/0282110	12/2015	Vagman et al.	N/A	N/A
2016/0282179	12/2015	Nazemi et al.	N/A	N/A
2016/0288331	12/2015	Sivich et al.	N/A	N/A
2016/0313114	12/2015	Tohme et al.	N/A	N/A
2016/0327383	12/2015	Becker et al.	N/A	N/A
2016/0340873	12/2015	Eidenberger et al.	N/A	N/A
2016/0341041	12/2015	Puura et al.	N/A	N/A
2016/0349746	12/2015	Grau	N/A	N/A
2016/0363436	12/2015	Clark et al.	N/A	N/A
2016/0363659	12/2015	Mindell et al.	N/A	N/A
2016/0363663	12/2015	Mindell et al.	N/A	N/A
2016/0363664	12/2015	Mindell et al.	N/A	N/A
2016/0364869	12/2015	Siercks et al.	N/A	N/A
2016/0364874	12/2015	Tohme et al.	N/A	N/A
2017/0066157	12/2016	Peters et al.	N/A	N/A
2017/0067739	12/2016	Siercks et al.	N/A	N/A
2017/0082436	12/2016	Siercks et al.	N/A	N/A
2017/0091922	12/2016	Siercks et al.	N/A	N/A
2017/0091923	12/2016	Siercks et al.	N/A	N/A
2017/0108528	12/2016	Atlas et al.	N/A	N/A
2017/0122733	12/2016	Brown	N/A	N/A
2017/0122736	12/2016	Dold et al.	N/A	N/A
2017/0166399	12/2016	Stubbs	N/A	N/A
2017/0173796	12/2016	Kim et al.	N/A	N/A
2017/0176572	12/2016	Charvat et al.	N/A	N/A
2017/0179570	12/2016	Charvat	N/A	N/A
2017/0179603	12/2016	Charvat et al.	N/A	N/A
2017/0191822	12/2016	Becker et al.	N/A	N/A
2017/0227355	12/2016	Pettersson et al.	N/A	N/A
2017/0236299	12/2016	Valkenburg et al.	N/A	N/A
2017/0254102	12/2016	Peters et al.	N/A	N/A
2017/0269203	12/2016	Trishaun	N/A	N/A
2017/0307757	12/2016	Hinderling et al.	N/A	N/A
2017/0314909	12/2016	Dang	N/A	N/A
2017/0314918	12/2016	Shah	N/A	N/A
2017/0333137	12/2016	Roessler	N/A	N/A
2017/0343336	12/2016	Lettau	N/A	N/A
2018/0003493	12/2017	Bernhard et al.	N/A	N/A
2018/0017384	12/2017	Siercks et al.	N/A	N/A
2018/0023935	12/2017	Atwell et al.	N/A	N/A
2018/0038684	12/2017	Fröhlich et al.	N/A	N/A
2018/0046096	12/2017	Shibazaki	N/A	N/A
2018/0052233	12/2017	Frank et al.	N/A	N/A
2018/0108178	12/2017	Murugappan et al.	N/A	N/A
2018/0121571	12/2017	Tiwari et al.	N/A	N/A

2018/0149469	12/2017	Becker et al.	N/A	N/A
2018/0156601	12/2017	Pontai	N/A	N/A
2018/0170719	12/2017	Tasch et al.	N/A	N/A
2018/0180416	12/2017	Edelman et al.	N/A	N/A
2018/0202796	12/2017	Ziegenbein	N/A	N/A
2018/0209156	12/2017	Pettersson	N/A	N/A
2018/0239010	12/2017	Mindell et al.	N/A	N/A
2018/0300433	12/2017	Maxam et al.	N/A	N/A
2019/0026401	12/2018	Benjamin et al.	N/A	N/A
2019/0032348	12/2018	Parkes	N/A	N/A
2019/0184555	12/2018	Linnell et al.	N/A	N/A
2019/0224846	12/2018	Pivac et al.	N/A	N/A
2019/0251210	12/2018	Pivac et al.	N/A	N/A
2019/0316369	12/2018	Pivac	N/A	G05B 19/416
2020/0173777	12/2019	Pivac et al.	N/A	N/A
2020/0206923	12/2019	Pivac et al.	N/A	N/A
2020/0206924	12/2019	Pivac et al.	N/A	N/A
2020/0215688	12/2019	Pivac et al.	N/A	N/A
2020/0215692	12/2019	Pivac et al.	N/A	N/A
2020/0215693	12/2019	Pivac et al.	N/A	N/A
2021/0016437	12/2020	Pivac et al.	N/A	N/A
2021/0016438	12/2020	Pivac et al.	N/A	N/A
2021/0080582	12/2020	Pivac et al.	N/A	N/A
2021/0291362	12/2020	Pivac et al.	N/A	N/A
2021/0370509	12/2020	Pivac et al.	N/A	N/A
2021/0379775	12/2020	Pivac et al.	N/A	N/A
2021/0404195	12/2020	Huth	N/A	B33Y 80/00
2022/0058300	12/2021	Pivac et al.	N/A	N/A
2023/0167650	12/2022	Pickering et al.	N/A	N/A
2024/0328180	12/2023	Pickering	N/A	E04G 21/20

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
645640	12/1993	AU	N/A
673498	12/1989	CH	N/A
2730976	12/2004	CN	N/A
2902981	12/2006	CN	N/A
2923903	12/2006	CN	N/A
101100903	12/2007	CN	N/A
201184054	12/2008	CN	N/A
101360873	12/2008	CN	N/A
101476883	12/2008	CN	N/A
100557169	12/2008	CN	N/A
101694130	12/2009	CN	N/A
201972413	12/2010	CN	N/A
102359282	12/2011	CN	N/A
202248944	12/2011	CN	N/A
202292752	12/2011	CN	N/A

102995911	12/2012	CN	N/A
202925913	12/2012	CN	N/A
103363902	12/2012	CN	N/A
103698769	12/2013	CN	N/A
203701626	12/2013	CN	N/A
104141391	12/2013	CN	N/A
104153591	12/2013	CN	N/A
104493810	12/2014	CN	N/A
204295678	12/2014	CN	N/A
104612411	12/2014	CN	N/A
204311767	12/2014	CN	N/A
103774859	12/2014	CN	N/A
103753586	12/2014	CN	N/A
105113373	12/2014	CN	N/A
105178616	12/2014	CN	N/A
105257008	12/2015	CN	N/A
105544998	12/2015	CN	N/A
205290958	12/2015	CN	N/A
104806028	12/2015	CN	N/A
205668271	12/2015	CN	N/A
205840368	12/2015	CN	N/A
205990775	12/2016	CN	N/A
206185879	12/2016	CN	N/A
206189878	12/2016	CN	N/A
105089274	12/2016	CN	N/A
105064699	12/2016	CN	N/A
107217859	12/2016	CN	N/A
107237483	12/2016	CN	N/A
107357294	12/2016	CN	N/A
107605167	12/2017	CN	N/A
206838382	12/2017	CN	N/A
206844687	12/2017	CN	N/A
107654077	12/2017	CN	N/A
107675891	12/2017	CN	N/A
107740591	12/2017	CN	N/A
106088632	12/2017	CN	N/A
107762165	12/2017	CN	N/A
207063553	12/2017	CN	N/A
106088631	12/2017	CN	N/A
107975245	12/2017	CN	N/A
108061551	12/2017	CN	N/A
108222527	12/2017	CN	N/A
108301628	12/2017	CN	N/A
108331362	12/2017	CN	N/A
106150109	12/2017	CN	N/A
108396977	12/2017	CN	N/A
108457479	12/2017	CN	N/A
108708560	12/2017	CN	N/A
208023979	12/2017	CN	N/A
106881711	12/2018	CN	N/A

107083845	12/2018	CN	N/A
108016585	12/2018	CN	N/A
209701519	12/2018	CN	N/A
3430915	12/1985	DE	N/A
4038260	12/1990	DE	N/A
4207384	12/1992	DE	N/A
19509809	12/1994	DE	N/A
4417928	12/1994	DE	N/A
29601535	12/1996	DE	N/A
19600006	12/1996	DE	N/A
19603234	12/1996	DE	N/A
19743717	12/1998	DE	N/A
19849720	12/1999	DE	N/A
10230021	12/2002	DE	N/A
102006030130	12/2006	DE	N/A
102009018070	12/2009	DE	N/A
102009042014	12/2010	DE	N/A
202012100646	12/2012	DE	N/A
102013019869	12/2014	DE	N/A
190076	12/1985	EP	N/A
370682	12/1989	EP	N/A
456020	12/1994	EP	N/A
493020	12/1994	EP	N/A
495525	12/1994	EP	N/A
836664	12/1998	EP	N/A
674069	12/1998	EP	N/A
1918478	12/2007	EP	N/A
2112291	12/2008	EP	N/A
2219528	12/2009	EP	N/A
2249997	12/2009	EP	N/A
2353801	12/2010	EP	N/A
2199719	12/2013	EP	N/A
3084719	12/2015	EP	N/A
2296556	12/2007	ES	N/A
2230825	12/1973	FR	N/A
2524522	12/1982	FR	N/A
119331	12/1917	GB	N/A
2198105	12/1922	GB	N/A
673472	12/1951	GB	N/A
682010	12/1951	GB	N/A
839253	12/1959	GB	N/A
1067604	12/1966	GB	N/A
1465068	12/1976	GB	N/A
2268536	12/1993	GB	N/A
125079	12/2000	GB	N/A
2422400	12/2005	GB	N/A
64006719	12/1988	JP	N/A
H07101509	12/1998	JP	N/A
2000127077	12/1999	JP	N/A
2005283600	12/2004	JP	N/A

4294990	12/2008	JP	N/A
2009521630	12/2008	JP	N/A
5508895	12/2013	JP	N/A
87054	12/1988	LU	N/A
87381	12/1989	LU	N/A
88144	12/1993	LU	N/A
85392	12/2008	RU	N/A
9702397	12/1996	WO	N/A
2001076830	12/2000	WO	N/A
2004020760	12/2003	WO	N/A
2004083540	12/2003	WO	N/A
2005014240	12/2004	WO	N/A
2005017550	12/2004	WO	N/A
2005070657	12/2004	WO	N/A
2004011734	12/2004	WO	N/A
2006111827	12/2005	WO	N/A
2007076581	12/2006	WO	N/A
2008124713	12/2007	WO	N/A
2009026641	12/2008	WO	N/A
2009026642	12/2008	WO	N/A
2009044002	12/2008	WO	N/A
2010020457	12/2009	WO	N/A
2011077006	12/2010	WO	N/A
2013088154	12/2012	WO	N/A
2013134559	12/2012	WO	N/A
2017162630	12/2016	WO	N/A
2018009978	12/2017	WO	N/A
2018009981	12/2017	WO	N/A
2018009985	12/2017	WO	N/A
2018009986	12/2017	WO	N/A
WO-2018009980	12/2017	WO	B25J 15/026
2018052469	12/2017	WO	N/A
2018099323	12/2017	WO	N/A
2019006511	12/2018	WO	N/A
2019014701	12/2018	WO	N/A
2019014702	12/2018	WO	N/A
2019014705	12/2018	WO	N/A
2019014706	12/2018	WO	N/A
2019014707	12/2018	WO	N/A
2019033165	12/2018	WO	N/A
2019033166	12/2018	WO	N/A
2019033170	12/2018	WO	N/A
2019068128	12/2018	WO	N/A
2019071313	12/2018	WO	N/A
2020014737	12/2019	WO	N/A
2020047574	12/2019	WO	N/A
WO-2020136563	12/2019	WO	B25J 11/00
2020210863	12/2019	WO	N/A
WO-2020193601	12/2019	WO	B33Y 10/00

OTHER PUBLICATIONS

Boston Dynamics: “Introducing Spot (previously SpotMini)”, Jun. 28, 2016, YouTube video, 1 page (screenshot of video); video retrieved at <<https://www.youtube.com/watch?v=tf7IEVTDjng>>. cited by applicant

Delgado, R. et al.: “Development and Control of an Omnidirectional Mobile Robot on an EtherCAT Network”, International Journal of Applied Engineering Research, vol. 11, No. 21, 2016, pp. 10586-10592, XP055574484. cited by applicant

Dorfler, K. et al.: “Mobile Robotic Brickwork ', Automation of a Discrete Robotic Fabrication Process Using an Autonomous Mobile Robot Robotic Fabrication in Architecture”, Art and Design 2016, Feb. 4, 2016 (Feb. 4, 2016), pp. 204-217, XP055567451. cited by applicant

Egerstedt, M. et al.: “Control of Mobile Platforms using a Virtual Vehicle Approach”, IEEE Transactions on Automatic Control, vol. 46, No. 11, Nov. 2001 (Nov. 1, 2001), XP055567515. cited by applicant

Examination Report mailed Apr. 18, 2021 in GCC Patent Application No. 2018-35644, 5 pages. cited by applicant

Examination Report mailed Apr. 30, 2021 in GCC Patent Application No. 2018-35643, 3 pages. cited by applicant

Examination Report mailed Jun. 29, 2021 for India Patent Application No. 201927004006, 6 pages. cited by applicant

Examination Report mailed Sep. 30, 2021 for Australian Patent Application No. 2017295316, 3 pages. cited by applicant

Extended European Search Report mailed Jun. 4, 2021 for European Patent Application No. 18865644.1, 7 pages. cited by applicant

Extended European Search Report mailed Mar. 16, 2021 for European Patent Application No. 18834565.6, 19 pages. cited by applicant

Extended European Search Report mailed Mar. 17, 2021 for European Patent Application No. 18835861.8, 12 pages. cited by applicant

Extended European Search Report mailed Mar. 18, 2021 for European Patent Application No. 18834673.8, 14 pages. cited by applicant

Extended European Search Report mailed Mar. 18, 2021 for European Patent Application No. 18834893.2, 12 pages. cited by applicant

Extended European Search Report mailed Mar. 18, 2021 for European Patent Application No. 18835737.0, 10 pages. cited by applicant

Extended European Search Report mailed Mar. 30, 2021 for European Patent Application No. 18845794.9, 13 pages. cited by applicant

Extended European Search Report mailed Mar. 5, 2021 for European Patent Application No. 18828425.1, 7 pages. cited by applicant

Fastbrick Robotics, Fastbrick Robotics: Hadrian 105 First Look Revealed, Nov. 16, 2015 (Nov. 16, 2015), XP054978174, Retrieved from the Internet <[URL:https://www.youtube.com/watch?v=7Zw7qHxMtrY](https://www.youtube.com/watch?v=7Zw7qHxMtrY)> [retrieved on Nov. 16, 2015]. cited by applicant

Fastbrick Robotics: Hadrian 105 Demonstrative Model Animation, Jun. 29, 2015 (Jun. 29, 2015), XP054979424, Retrieved from the Internet <[URL:https://www.youtube.com/watch?v=Rebqcsb61gY](https://www.youtube.com/watch?v=Rebqcsb61gY)> [retrieved on Mar. 7, 2018]. cited by applicant

Fastbrick Robotics: Hadrian 105 Time Lapse, Fastbrick Robotics Time Lapse, May 22, 2016 (May 22, 2016), XP054978173, Retrieved from the Internet <[URL:https://www.youtube.com/watch?v=4YcrO8ONcfY](https://www.youtube.com/watch?v=4YcrO8ONcfY)> [retrieved on May 22, 2016]. cited by applicant

Fastbrick Robotics: Hadrian X Digital Construction System, published on Sep. 21, 2016 <[URL:https://www.youtube.com/watch?v=5bW1vuCgEaA](https://www.youtube.com/watch?v=5bW1vuCgEaA)>. cited by applicant

Feng, C. et al.: “Vision Guided Autonomous Robotic Assembly and as-built Scanning on

Unstructured Construction Sites”, Automation in Construction, vol. 59, Nov. 2015 (Nov. 1, 2015), pp. 128-138, XP055567454. cited by applicant

Gander H et al: “Application of a floating point digital signal processor to a dynamic robot measurement system”, Instrumentation and Measurement Technology Conference, 1994. IMTC/94. Conference Proceedings. 10.SUP.th .Anniversary. Advanced Technologies in I & M., 1994 IEEE Hamamatsu, Japan May 10-12, 1994, New York, NY, USA, IEEE, May 10, 1994 (May 10, 1994), pp. 372-375, XP010121924, DOI: 10.1109/IMTC.1994.352046, ISBN: 978-0-7803-1880-9, *whole document*. cited by applicant

Gao, X. et al.: “Complete Solution Classification for the Perspective-Three-Point Problem”, IEEE Transactions on Pattern Analysis and Machine Intelligence, vol. 25, No. 8, Aug. 2003 (Aug. 1, 2003), pp. 930-943, XP011099374. cited by applicant

Garrido, S. et al., “FM2: A real-time fast marching sensor based motion planner”, Advanced Intelligent Mechatronics, 2007 IEEE/ASME International Conference on, IEEE, PI, Sep. 1, 2007 (Sep. 1, 2007), pp. 1-6. cited by applicant

Gifftthaler, M. et al., “Efficient Kinematic Planning for Mobile Manipulators with Non-holonomic Constraints Using Optimal Control”, 2017 IEEE International Conference on Robotics and Automation (ICRA), Singapore, May 29-Jun. 3, 2017. cited by applicant

Heintze, H., “Design and Control of a Hydraulically Actuated Industrial Brick Laying Robot,” 264 pages. cited by applicant

Heintze, J. et al., “Controlled hydraulics for a direct drive brick laying robot,” Automation in Construction 5 (1996), pp. 23-29. cited by applicant

Helm, V. et al.: “Mobile Robotic Fabrication on Construction Sites: dimRob”, IEEE /RSJ International Conference on Intelligent Robots and Systems, Oct. 7, 2012 (Oct. 7, 2012), Vilamoura, Portugal, pp. 4335-4341, XP032287463. cited by applicant

<http://www.new-technologies.org/ECT/Other/brickrob.htm>. “Emerging Construction Technologies.” Dec. 1, 2006. cited by applicant

Huang, S. et al., “Applying High-Speed Vision Sensing to an Industrial Robot for High-Performance Position Regulation under Uncertainties,” Sensors, 2016, 16, 1195, 15 pages. cited by applicant

International Search Report and Written Opinion for International Application No. PCT/AU2021/050732; Date of Mailing: Sep. 21, 2021; 10 pages. cited by applicant

International Preliminary Report on Patentability for International Application No. PCT/AU2017/050731; Date of Mailing: Jan. 15, 2019; 5 pages. cited by applicant

International Preliminary Report on Patentability for International Application No. PCT/AU2017/050738; Date of Mailing: Jan. 15, 2019; 13 pages. cited by applicant

International Preliminary Report on Patentability for International Application No. PCT/AU2017/050739; Date of Mailing: Jan. 15, 2019; 6 pages. cited by applicant

International Preliminary Report on Patentability for International Application No. PCT/AU2018/050733; Date of Mailing: Jan. 21, 2020; 6 pages. cited by applicant

International Preliminary Report on Patentability for International Application No. PCT/AU2018/050734; Date of Mailing: Jan. 21, 2020; 9 pages. cited by applicant

International Preliminary Report on Patentability for International Application No. PCT/AU2018/050737; Date of Mailing: Jan. 21, 2020; 6 pages. cited by applicant

International Preliminary Report on Patentability for International Application No. PCT/AU2018/050739; Date of Mailing: Jan. 21, 2020; 6 pages. cited by applicant

International Preliminary Report on Patentability for International Application No. PCT/AU2018/050740; Date of Mailing: Jan. 21, 2020; 6 pages. cited by applicant

International Search Report and Written Opinion for International Application No. PCT/AU2017/050730; Date of Mailing: Aug. 23, 2017; 17 pages. cited by applicant

International Search Report and Written Opinion for International Application No.

PCT/AU2017/050731; Date of Mailing: Aug. 31, 2017; 8 pages. cited by applicant
International Search Report and Written Opinion for International Application No. PCT/AU2017/050738; Date of Mailing: Oct. 17, 2017; 19 pages. cited by applicant
International Search Report and Written Opinion for International Application No. PCT/AU2017/050739; Date of Mailing: Sep. 28, 2017; 9 pages. cited by applicant
International Search Report and Written Opinion for International Patent Application No. PCT/AU19/50742; Date of Mailing Sep. 23, 2019; 5 pages. cited by applicant
International Search Report and Written Opinion for International Patent Application No. PCT/AU19/50743; Date of Mailing mailed Oct. 1, 2019; 10 pages. cited by applicant
International Search Report and Written Opinion for International Patent Application No. PCT/AU20/50367; Date of Mailing Jun. 29, 2020; 15 pages. cited by applicant
International Search Report and Written Opinion for International Patent Application No. PCT/AU20/50368; Date of Mailing Jun. 25, 2020; 11 pages. cited by applicant
Kazemi, M. et al.: "Path Planning for Image-based Control of Wheeled Mobile Manipulators", 2012 IEEE /RSJ International Conference on Intelligent Robots and Systems, Oct. 7, 2012 (Oct. 7, 2012), Vilamoura, Portugal, XP055567470. cited by applicant
Kleinigger, M. et al: "Application of 6-DOF sensing for robotic disturbance compensation", Automation Science and Engineering (CASE), 2010 IEEE Conference on, IEEE, Piscataway, NJ, USA, Aug. 21, 2010 (Aug. 21, 2010, pp. 344-349, XP031762876, ISBN: 978-1-4244-5477-1, *abstract*, *sections 1 to 3*. cited by applicant
Kleinkes, M. et al.: "Laser Tracker and 6DoF measurement strategies in industrial robot applications", CMSC 2011: Coordinate Metrology System Conference, Jul. 25, 2011 (Jul. 25, 2011), XP055456272. cited by applicant
Koren et al.: "End-effector guidance of robot arms", CIRP Annals-Manufacturing Technology, vol. 36, No. 1, 1987, pp. 289-292, XP055456270. cited by applicant
Kwon, S. et al., "On the Coarse/Fine Dual-Stage Manipulators with Robust Perturbation Compensator," IEEE, May 21-26, 2001, pp. 121-126. cited by applicant
Kyle in CMSC: Charlotte-Concord, Jul. 21-25, 2008. cited by applicant
Latteur, et al., "Drone-Based Additive Manufacturing of Architectural Structures," IASS Symposium 2015, Amsterdam, The Netherlands; Aug. 17-20, 2015; 12 pages. cited by applicant
Lippiello, V. et al.: "Position-Based Visual Servoing in Industrial Multirobot Cells Using a Hybrid Camera Configuration", IEEE Transactions on Robotics, vol. 23, No. 1, Feb. 2007 (Feb. 1, 2007), XP011163518. cited by applicant
Liu, Z. et al.: "EtherCAT Based Robot Modular Joint Controller", Proceeding of the 2015 IEEE International Conference on Information and Automation, Aug. 2015 (Aug. 1, 2015), Lijiang, China, pp. 1708-1713, XP033222650. cited by applicant
Mercedes-Benz: "Mercedes-Benz "Chicken" Magic Body Control TV commercial", YouTube, Sep. 23, 2013, 1 page. Retrieved from the internet: <<https://www.youtube.com/watch?v+nLwML2PagbY>>. cited by applicant
Notice of Acceptance of Patent Application received for priority Australian Patent Application No. 2017294796, mailed May 15, 2019 (158 pages). cited by applicant
Office Action mailed Apr. 21, 2021 in Japanese Patent Application No. 2019-523148, 4 pages. cited by applicant
Office Action mailed Aug. 20, 2021 for Japanese Patent Application No. 2019-523147, 3 pages. cited by applicant
Office Action mailed Jul. 5, 2021 for Japanese Patent Application No. 2019-523145, 4 pages. cited by applicant
Office Action mailed May 24, 2021 for Chinese Patent Application No. 201880067520.0, 8 pages. cited by applicant
Office Action mailed Sep. 3, 2021 for Chinese Patent Application No. 201780056460.8, 9 pages.

cited by applicant

Partial Supplementary European Search Report mailed Apr. 14, 2020 in European Patent Application No. 17826696.1, 10 pages. cited by applicant

Pless, R. : “Using Many Cameras as One”, IEEE Computer Society Conference on Computer Vision and Pattern Recognition, Jun. 18, 2003 (Jun. 18, 2003), Madison , WI, USA, pp. 1-7, XP055564465. cited by applicant

Posada et al.: “High accurate robotic drilling with external sensor and compliance model-based compensation”, Robotics and Automation (ICRA), 2016 IEEE International Conference, May 16, 2016 (May 16, 2016), pp. 3901-3907, XP032908649. cited by applicant

Pritschow, G. et al., “A Mobile Robot for On-Site Construction of Masonry,” Inst. of Control Tech. for Machine Tools and Manuf. Units, pp. 1701-1707. cited by applicant

Pritschow, G. et al., “Application Specific Realisation of a Mobile Robot for On-Site Construction of Masonry,” Automation and Robotics in Construction XI, 1994, pp. 95-102. cited by applicant

Pritschow, G. et al., “Configurable Control System of a Mobile Robot for On-Site Construction of Masonry,” Inst. of Control Technology for Machine Tools and Manuf. Units, pp. 85-92. cited by applicant

Pritschow, G. et al., “Technological aspects in the development of a mobile bricklaying robot,” Automation in Construction 5 (1996), pp. 3-13. cited by applicant

Riegl Laser Measurement Systems. “Long Range & High Accuracy 3D Terrestrial Laser Scanner System—LMS-Z420i.” pp. 1-4. cited by applicant

Salcudean, S. et al., “On the Control of Redundant Coarse-Fine Manipulators,” IEEE, pp. 1834-1840. cited by applicant

Sandy, T. et al.: “Autonomous Repositioning and Localization of an In Situ Fabricator”, 2016 IEEE International Conference on Robotics and Automation (ICRA), May 16, 2016 (May 16, 2016), pp. 2852-2858, XP055567467. cited by applicant

Siciliano, B. et al., “Robotics—chapters 2-4” Robotics, Dec. 31, 2009 (Dec. 31, 2009), Springer London, London, pp. 39-189. cited by applicant

Skibniewski, M.J., “Current Status of Construction Automation and Robotics in the United States of America,” The 9th International Symposium on Automation and Robotics in Construction, Jun. 3-5, 1992, 8 pages. cited by applicant

Trimble ATS. “Advanced Tracking Sensor (ATS) with target recognition capability for stakeless machine control survey applications.” pp. 1-4. cited by applicant

Vincze, M. et al., “A Laser Tracking System to Measure Position and Orientation of Robot End Effectors Under Motion,” The International Journal of Robotics Research, vol. 13, No. 4, Aug. 1994, pp. 305-314. cited by applicant

Warszawski, A. et al., “Implementation of Robotics in Building: Current Status and Future Prospects,” Journal of Construction Engineering and Management, Jan./Feb. 1998, 124(1), pp. 31-41. cited by applicant

Willmann, J. et al.: “Robotic Timber Construction—Expanding Additive Fabrication to New Dimensions”, Automation in Construction, vol. 61, 2016, pp. 16-23, XP029310896. cited by applicant

Xu, H. et al.: “Uncalibrated Visual Servoing of Mobile Manipulators with an Eye-to-hand Camera”, Proceedings of the 2016 IEEE International Conference on Robotics and Biomimetics, Dec. 3, 2016 (Dec. 3, 2016), Qingdao, China, pp. 2145-2150, XP033071767. cited by applicant

Yu, S.N. et al., “Feasibility verification of brick-laying robot using manipulation trajectory and the laying pattern optimization,” Dept. of Mech. Eng., Automation in Construction (2009), pp. 644-655. cited by applicant

Zaki, T., “Parametric modeling of Blackwall assemblies for automated generation of shop drawings and detailed estimates using BIM”, Master's Thesis, May 23, 2016, pp. 1-151. cited by applicant
Examination report dated Sep. 13, 2022 on Chinese Patent Application No. 201880066756.2, 11

pages. cited by applicant
IPRP dated Jan. 10, 2023 on International Patent Application No. PCT/AU2021/050732, 5 pages.
cited by applicant
Examination report dated Sep. 30, 2023 on Australian Patent Application No. AU2018348785.
cited by applicant
Examination report dated Sep. 30, 2023 on UAE Patent Application No. P60005242020. cited by
applicant
Examination report issued Mar. 31, 2023 on Saudi Arabian Patent Application No. 522440994.
cited by applicant
Examination Report dated Jul. 4, 2024 on European Patent Application No. 21837579.8. cited by
applicant
International Preliminary Report on Patentability received for counterpart International Application
No. PCT/AU2023/050325, mailed Oct. 8, 2024, 6 pages. cited by applicant
International Preliminary Report on Patentability received for International Patent Application No.
PCT/AU2021/050361, mailed Oct. 25, 2022. cited by applicant

Primary Examiner: Ference; James M

Attorney, Agent or Firm: Perkins Coie LLP

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

(1) The present application is a United States national phase entry of International Application No. PCT/AU2021/050732, titled “ADHESIVE APPLICATION SYSTEM” and filed on Jul. 8, 2021 which claims priority from Australian Provisional Application No. 2020902367 titled “ADHESIVE APPLICATION SYSTEM” and filed on 8 Jul. 2020, the contents of which are hereby incorporated by reference in their entireties.

TECHNICAL FIELD

(2) The present disclosure relates to an adhesive application system for an automated block laying machine. In a particular form, the system may be used to apply adhesive onto a brick or block, however the system may also be used to apply adhesive onto other construction materials such as roof tiles and the like.

BACKGROUND

(3) The reference in this specification to any prior publication (or information derived from it), or to any matter which is known, is not, and should not be taken as an acknowledgment or admission or any form of suggestion that the prior publication (or information derived from it) or known matter forms part of the common general knowledge in the field of endeavour to which this specification relates.

(4) WO 2007/076581 describes an automated brick laying system for constructing a building from a plurality of bricks comprising a robot provided with a brick laying and adhesive applying head, a measuring system, and a controller that provides control data to the robot to lay the bricks at predetermined locations. The measuring system measures in real time the position of the head and produces position data for the controller. The controller produces control data on the basis of a comparison between the position data and a predetermined or pre-programmed position of the head to lay a brick at a predetermined position for the building under construction. The controller can control the robot to construct the building in a course by course manner where the bricks are laid sequentially at their respective predetermined positions and where a complete course of bricks for

the entire building is laid prior to laying of the bricks for the next course.

(5) A number of adhesive application systems suitable for use with robotic construction equipment have been contemplated by the present applicant. For example in PCT/AU2019/000102, there is provided an adhesive application system for an automated brick laying machine, the adhesive application system comprising an adhesive reservoir, an adhesive supply line extending from the reservoir to one or more nozzles configured to dispense adhesive, a first pump (a primary pump) positioned near to the reservoir, at least one further pump (secondary pump) positioned near to the or each of the nozzles, a supply line load cell positioned near to the or each secondary pump, and a control system, wherein in use, the control system controls operation of the primary pump in response to a supply pressure sensed at the supply line load cell. An adhesive applicator assembly is provided comprising a base, at least one nozzle carriage assembly, each nozzle carriage assembly comprising a carriage, and at least one adhesive outlet nozzle depending from the carriage, for dispensing adhesive received from a supply of adhesive via an adhesive supply line, and a transverse drive means for driving the or each nozzle carriage assembly transversely relative to the base. The above described arrangement is a top-down pump system in which adhesive is pumped down onto an upward facing side of a brick so that gravity assists in applying adhesive to the brick.

(6) The adhesive application system described in PCT/AU2019/000102 including the adhesive supply is co-located at the brick laying and adhesive applying head. This arrangement is in contrast to the system described in PCT/AU2017/050728, in which an adhesive container and pump was located proximate the chassis of a truck that provides the base of the automated brick laying machine and the adhesive was pumped out along the boom to an adhesive applicator mounted to the brick laying and adhesive applying head. The adhesive applicator in this system comprised an applicator head with nozzles supported on a distal end of a tongue member, said tongue member being housed in a sheath for linear movement to extend said adhesive applicator head across the surface of a brick for application of adhesive.

(7) It is desirable to develop an adhesive application system for an automated block laying machine that provides an alternative to a top-down pumping system. In a top-down pumping system, dispensing adhesive in a metered dose can be difficult due to varying viscosity of the adhesive with temperature. As adhesive is applied to a top face of the block, it must be rotated before it is laid which can induce the adhesive to drip from the block. Furthermore, applying adhesive with nozzles can be difficult on blocks with complex grid/core structures. It would also be advantageous to reduce design complexity, part count and running costs whilst increasing adhesive application speed.

(8) It is against this background and the problems and difficulties associated therewith that the present invention has been developed.

(9) Other objects and advantages of the present invention will become apparent from the following description, taken in connection with the accompanying drawings, wherein, by way of illustration and example, at least one embodiment of the present invention is disclosed.

SUMMARY

(10) In one broad form, an aspect of the present invention seeks to provide an adhesive application system for an automated block laying machine, the adhesive application system including: a) an adhesive supply; and, b) an adhesive applicator including: i) a container for storing an adhesive reservoir received from the adhesive supply via an adhesive supply line; ii) a lid mounted to the container so as to substantially cover the adhesive reservoir; and, iii) an adhesive roller rotatably coupled to the container and configured to be at least partially submerged in the adhesive reservoir so that upon rotation adhesive from the reservoir is entrained up and around at least a portion of the adhesive roller, wherein, in use, adhesive is transferred from the adhesive roller onto a bottom surface of a block as the block is moved across the roller.

(11) In one embodiment, adhesive is provided to the adhesive applicator from the adhesive supply via a gravity feed.

- (12) In one embodiment, the adhesive supply is a canister having: a) an opening to allow the canister to be at least partially filled with adhesive; and, b) a removable cap that seals the opening and is configured to be in fluid communication with the adhesive supply line; wherein, in use, the canister is oriented so that the cap is facing downward and adhesive is able to flow by gravity through the cap and into the supply line.
- (13) In one embodiment, the adhesive supply line includes a valve which is openable to allow flow of adhesive into the container of the adhesive applicator.
- (14) In one embodiment, the valve is actuated by a motor and controllable to selectively open in order to meter the amount of adhesive supplied to the container.
- (15) In one embodiment, the motor is coupled to a reduction gearbox to increase torque for actuating the valve.
- (16) In one embodiment, the amount of adhesive in the container is monitored by at least one of: a) a level sensor that provides a signal to a controller indicative of the level of adhesive in the container; and, b) a mass sensor that provides a signal to a controller indicative of the mass of the container.
- (17) In one embodiment, the level sensor is a capacitive sensor mounted to the side of the container.
- (18) In one embodiment, the mass sensor is a load cell mounted beneath the container.
- (19) In one embodiment, the adhesive is heated prior to application.
- (20) In one embodiment, a heating element is provided at the base of the container.
- (21) In one embodiment, the adhesive roller includes: a) an elongate shaft having axles rotatably coupled to opposing ends of the container; and, b) one or more roller elements coupled to or integral with the shaft.
- (22) In one embodiment, in use, the one or more roller elements are at least partially submerged in the adhesive reservoir.
- (23) In one embodiment, the one or more roller elements at least partially protrude above the lid through openings therein.
- (24) In one embodiment, the one or more roller elements are circular discs or wheels.
- (25) In one embodiment, the one or more roller elements have at least one circumferential groove formed therein.
- (26) In one embodiment, the adhesive roller is driven by a motor coupled to the elongate shaft via a coupling element.
- (27) In one embodiment, a bead of adhesive is entrained onto the one or more roller elements as the shaft rotates and is transferred onto the block as the block moves across the adhesive roller, wherein one of: a) the block contacts the one or more roller elements; and, b) there is a gap between the block and the one or more roller elements less than or equal to the thickness of the bead of adhesive.
- (28) In one embodiment, the system is mounted to a block laying and adhesive applying head of the automated block laying machine, the head located at a distal end of a telescopingly extendable and foldable boom for conveying blocks therealong.
- (29) In one embodiment, the adhesive applicator is adjustably mounted to the head via a support post and bracket.
- (30) In another broad form, an aspect of the present invention seeks to provide a block laying and adhesive applying head for an automated block laying machine, the head mountable to a distal end of a telescopingly extendable and foldable boom for conveying blocks therealong, the head including an adhesive application system including: a) an adhesive supply; and, b) an adhesive applicator including: i) a container for storing an adhesive reservoir received from the adhesive supply via an adhesive supply line; ii) a lid mounted to the container so as to substantially cover the adhesive reservoir; and, iii) an adhesive roller rotatably coupled to the container and configured to be at least partially submerged in the adhesive reservoir so that upon rotation adhesive from the

reservoir is entrained up and around at least a portion of the adhesive roller, wherein, in use, adhesive is transferred from the adhesive roller onto a bottom surface of a block as the block is moved across the roller.

(31) In one embodiment, the head includes a block transfer apparatus for transferring a block from a distal end of the boom to a robotic arm having an end effector which grips and lays the block at a predetermined spatial position and orientation, the block transfer apparatus configured to clamp the block and linearly extend it across the adhesive roller before it is transferred to the robotic arm and laid.

(32) In another broad form, an aspect of the present invention seeks to provide a block laying and adhesive applying head for an automated block laying machine, the head mountable to a distal end of a telescopingly extendable and foldable boom for conveying blocks therealong, the head including: a) a support mountable to the boom; b) a robotic arm having an end effector operable to grip and lay blocks at a predetermined spatial position and orientation; c) a block transfer apparatus for transferring a block from the distal end of the boom to the robotic arm; and, d) an adhesive application system mounted to the support including: i) an adhesive supply; and, ii) an adhesive applicator including: (1) a container for storing an adhesive reservoir received from the adhesive supply via an adhesive supply line; (2) a lid mounted to the container so as to substantially cover the adhesive reservoir; and, (3) an adhesive roller rotatably coupled to the container and configured to be at least partially submerged in the adhesive reservoir so that upon rotation adhesive from the reservoir is entrained up and around at least a portion of the adhesive roller, wherein, in use, adhesive is transferred from the adhesive roller onto a bottom surface of a block as the block transfer apparatus linearly extends a block across the roller.

(33) In yet a further broad form, an aspect of the present invention seeks to provide an automated block laying machine, including: a) a base; b) a telescopingly extendable and foldable boom rotatably mounted from the base for conveying blocks therealong; and, c) a block laying and adhesive applying head, the head mountable to a distal end of the boom, the head including: i) a support mountable to the boom; ii) a robotic arm having an end effector operable to grip and lay blocks at a predetermined spatial position and orientation; iii) a block transfer apparatus for transferring a block from the distal end of the boom to the robotic arm; and, iv) an adhesive application system mounted to the support including: (1) an adhesive supply; and, (2) an adhesive applicator including: (a) a container for storing an adhesive reservoir received from the adhesive supply via an adhesive supply line; (b) a lid mounted to the container so as to substantially cover the adhesive reservoir; and, (c) an adhesive roller rotatably coupled to the container and configured to be at least partially submerged in the adhesive reservoir so that upon rotation adhesive from the reservoir is entrained up and around at least a portion of the adhesive roller, wherein, in use, adhesive is transferred from the adhesive roller onto a bottom surface of a block as the block transfer apparatus linearly extends a block across the roller.

(34) In another broad form, an aspect of the present invention seeks to provide an adhesive application system for use in an automated block laying machine, the adhesive application system including: a) an adhesive supply; and, b) an adhesive applicator including: i) a container for storing an adhesive reservoir received from the adhesive supply via an adhesive supply line; ii) a pair of sprung roller arms coupled to the container for supporting an adhesive roller; and, iii) an adhesive roller rotatably coupled to the sprung roller arms and configured to extend across the container, the adhesive roller at least partially submerged in the adhesive reservoir so that upon rotation adhesive from the reservoir is entrained up and around the roller; wherein, in use, adhesive is transferred from the adhesive roller onto a bottom surface of a block as the block is moved across the roller.

(35) In one embodiment, the pair of sprung roller arms form part of a lid that at least partly conceals the adhesive reservoir in the container.

(36) In one embodiment, the pair of sprung roller arms are disposed about opposing sides of the container.

- (37) In one embodiment, the pair of sprung roller arms include a groove or slot into which an axle of the adhesive roller is rotatably coupled.
- (38) In one embodiment, the sprung roller arms are coupled to the container by an integral moulding that includes a built-in hinge or pivot that allows the arms to flex.
- (39) In one embodiment, the adhesive roller has an elongate body having a plurality of circumferential grooves spaced apart along the length of the body.
- (40) In one embodiment, the adhesive roller includes: a) an elongate shaft; and, b) a plurality of spaced apart roller elements coupled to the shaft.
- (41) In one embodiment, the roller elements are cylindrical and include a plurality of circumferential grooves.
- (42) In one embodiment, the roller elements are spaced apart by spacer elements.
- (43) In one embodiment, the adhesive roller includes alternating roller and spacer elements coupled to the shaft and wherein the spacer elements have a diameter less than the roller elements.
- (44) In one embodiment, the adhesive applicator further includes a scraper coupled to the pair of sprung roller arms proximate a distal end thereof, the scraper having an elongate body defining: a) one or more ridged sections including a plurality of ridges operable to block particular grooves of the adhesive roller and scrape adhesive therefrom; and, b) one or more voids positioned between the ridged sections, the voids providing areas where adhesive is able to be entrained onto the roller to thereby define a pre-defined adhesive signature.
- (45) In one embodiment, the scraper is removably mounted to the roller arms.
- (46) In one embodiment, the scraper is urged beneath tabs located along a lengthwise edge of the lid so as to pre-tension the pair of sprung roller arms.
- (47) In one embodiment, the container is mounted in a cradle.
- (48) In one embodiment, a heated plate is mounted to the base of the cradle and the container is mounted onto the heated plate so as to control the temperature of the adhesive in the reservoir.
- (49) In one embodiment, a load cell is positioned beneath the cradle and used to monitor the weight of the adhesive in the reservoir so as to maintain a programmed level of adhesive in the reservoir.
- (50) In one embodiment, the lid further includes a pivotable cover disposed between the pair of sprung roller arms and extending towards the adhesive roller, a distal end of the pivotable cover urged beneath tabs disposed on the sprung roller arms so as to allow any adhesive on the surface of the cover to drain back into the reservoir.
- (51) In one embodiment, adhesive from the adhesive supply is pumped into the container via a peristaltic pump.
- (52) In one embodiment, the pump is one of: a) servo controlled; and, b) a linear peristaltic pump comprised of multiple pinch valves.
- (53) In one embodiment, the system is mounted to a block laying and adhesive applying head of the automated block laying machine, the head located at a distal end of a telescopingly extendable and foldable boom for conveying blocks therealong.
- (54) In one embodiment, the adhesive applicator is adjustably mounted to the head via a support post and bracket.
- (55) In another broad form, an aspect of the present invention seeks to provide a block laying and adhesive applying head for an automated block laying machine, the head mountable to a distal end of a telescopingly extendable and foldable boom for conveying blocks therealong, the head including an adhesive application system including: a) a container for storing an adhesive reservoir received from the adhesive supply via an adhesive supply line; b) a pair of sprung roller arms coupled to the container for supporting an adhesive roller; and, c) an adhesive roller rotatably coupled to the sprung roller arms and configured to extend across the container, the adhesive roller at least partially submerged in the adhesive reservoir so that upon rotation adhesive from the reservoir is entrained up and around the roller; wherein, in use, adhesive is transferred from the adhesive roller onto a bottom surface of a block as the block is moved across the roller.

(56) In one embodiment, the head includes a block transfer apparatus for transferring a block from a distal end of the boom to a robotic arm having an end effector which grips and lays the block at a predetermined spatial position and orientation, the block transfer apparatus configured to clamp the block and linearly extend it across the adhesive roller before it is transferred to the robotic arm and laid.

(57) In another broad form, an aspect of the present invention seeks to provide a block laying and adhesive applying head for an automated block laying machine, the head mountable to a distal end of a telescopingly extendable and foldable boom for conveying blocks therealong, the head including: a) a support mountable to the boom; b) a robotic arm having an end effector operable to grip and lay blocks at a predetermined spatial position and orientation; c) a block transfer apparatus for transferring a block from the distal end of the boom to the robotic arm; and, d) an adhesive application system mounted to the support including: i) a container for storing an adhesive reservoir received from the adhesive supply via an adhesive supply line; ii) a pair of sprung roller arms coupled to the container for supporting an adhesive roller; iii) an adhesive roller rotatably coupled to the sprung roller arms and configured to extend across the container, the adhesive roller at least partially submerged in the adhesive reservoir so that upon rotation adhesive from the reservoir is entrained up and around the roller; wherein, in use, adhesive is transferred from the adhesive roller onto a bottom surface of a block as the block transfer apparatus linearly extends a block across the roller.

(58) In another broad form, an aspect of the present invention seeks to provide an automated block laying machine, including: a) a base; b) a telescopingly extendable and foldable boom rotatably mounted from the base for conveying blocks therealong; and, c) a block laying and adhesive applying head, the head mountable to a distal end of the, the head including: i) a support mountable to the boom; ii) a robotic arm having an end effector operable to grip and lay blocks at a predetermined spatial position and orientation; iii) a block transfer apparatus for transferring a block from the distal end of the boom to the robotic arm; and, iv) an adhesive application system mounted to the support including: (1) an adhesive supply; and, (2) an adhesive applicator, including: (a) a container for storing an adhesive reservoir received from the adhesive supply via an adhesive supply line; (b) a pair of sprung roller arms coupled to the container for supporting an adhesive roller; and, (c) an adhesive roller rotatably coupled to the sprung roller arms and configured to extend across the container, the adhesive roller at least partially submerged in the adhesive reservoir so that upon rotation adhesive from the reservoir is entrained up and around the roller; wherein, in use, adhesive is transferred from the adhesive roller onto a bottom surface of a block as the block transfer apparatus linearly extends a block across the roller.

(59) In another broad form, an aspect of the present invention seeks to provide an adhesive application system for an automated block laying machine, the adhesive application system including: a) an adhesive supply; b) an adhesive applicator including: i) a container for storing an adhesive reservoir received from the adhesive supply via an adhesive supply line; ii) a lid mounted to the container so as to substantially cover the adhesive reservoir; and, iii) an adhesive roller rotatably coupled to the lid so that upon rotation adhesive from the reservoir is entrained up and around the roller, wherein, in use, adhesive is transferred from the adhesive roller onto a bottom surface of a block as the block is moved across the roller.

(60) It will be appreciated that the broad forms of the invention and their respective features can be used in conjunction and/or independently, and reference to separate broad forms is not intended to be limiting. Furthermore, it will be appreciated that features of the method can be performed using the system or apparatus and that features of the system or apparatus can be implemented using the method.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) Various examples and embodiments of the present invention will now be described with reference to the accompanying drawings, in which:—
- (2) FIG. 1A is an upper perspective view of an example of an adhesive application system for an automated block laying machine;
- (3) FIG. 1B is a lower perspective view of the adhesive application system of FIG. 1A;
- (4) FIG. 1C is a top view of the adhesive application system of FIG. 1A;
- (5) FIG. 1D is a front view of the adhesive application system of FIG. 1A showing a block disposed above the applicator;
- (6) FIG. 1E is a cross-sectional view taken through section A-A of FIG. 1C showing the adhesive roller drive system;
- (7) FIG. 1F is a cross-sectional view taken through section B-B of FIG. 1C showing the adhesive supply line valve actuation system;
- (8) FIG. 2A is an upper perspective view of the adhesive application system of FIG. 1A showing a disposable trip tray and bump plate incorporated into the system;
- (9) FIG. 2B is an upper perspective view of the adhesive application system of FIG. 2A showing an outer enclosure fitted to the system;
- (10) FIGS. 3A and 3B provide respective side and perspective views of a block laying and adhesive applying head and part of a boom assembly of an automated block laying machine including the adhesive application system of FIG. 1A;
- (11) FIG. 4A is a perspective view of an example of an adhesive applicator of the adhesive application system of FIG. 1A;
- (12) FIG. 4B is an exploded view of the adhesive applicator of FIG. 4A;
- (13) FIG. 5A is a perspective view of an example of an adhesive application system for an automated block laying machine;
- (14) FIG. 5B is a cross-sectional view of the adhesive application system of FIG. 5A taken through a sprung roller arm of the adhesive applicator;
- (15) FIG. 6A is a front view of the adhesive applicator of FIG. 5A;
- (16) FIG. 6B is a perspective view of the adhesive applicator of FIG. 5A;
- (17) FIG. 6C is a top view of the adhesive applicator of FIG. 5A;
- (18) FIG. 7 is a top view of a scraper for use with the adhesive applicator;
- (19) FIG. 8 is a cross-sectional view of the adhesive applicator;
- (20) FIG. 9A is a lower perspective view of a lid of the adhesive applicator;
- (21) FIG. 9B is a cross-sectional view of the lid of FIG. 9A;
- (22) FIG. 9C is a cross-sectional view of the lid of FIG. 9A with drain element pushed down;
- (23) FIG. 9D is a cross-sectional view of the lid of FIG. 9A showing a sprung roller arm pushed down to provide a pre-load when the roller is in contact with the block;
- (24) FIG. 10A is a further perspective view of the adhesive application system of FIG. 5A;
- (25) FIG. 10B is a cross-sectional view of the adhesive application system taken through the load cell beneath the adhesive applicator;
- (26) FIGS. 11A to 11H provide a sequence of views illustrating a block moving across the roller to thereby apply adhesive to the bottom surface of the block;
- (27) FIGS. 12A to 12C show photographs of example adhesive signatures applied to a variety of block geometries;
- (28) FIGS. 13A and 13B are perspective views of an automated block laying machine showing a boom in folded and extended positions;
- (29) FIG. 14 is a perspective view of a block laying and adhesive applying head of an automated block laying machine including the adhesive application system of FIG. 1A;
- (30) FIGS. 15A to 15C illustrate a sequence of perspective views of the block laying and adhesive

applying head of FIG. 14 extending a block across the roller of the adhesive applicator;
(31) FIG. 16 is a schematic side view of an adhesive applicator having a self-priming system;
(32) FIG. 17A is a schematic top view of a further example of an adhesive roller;
(33) FIG. 17B is a schematic top view of a further example of an adhesive roller;
(34) FIG. 17C is a schematic top view of a further example of an adhesive roller;
(35) FIG. 17D is a schematic top view of a further example of an adhesive roller;
(36) FIG. 17E is a schematic top view of yet a further example of an adhesive roller; and,
(37) FIG. 17F is a schematic top view of the adhesive roller of FIG. 17E shown beneath a lid which masks the leading edge of the roller.

DETAILED DESCRIPTION

(38) An example of an adhesive application system **10** for an automated block laying machine will now be described with reference to FIG. 1A.

(39) In this example, the system **10** includes an adhesive supply **140** and an adhesive applicator **100** including a container **110** for storing an adhesive reservoir received from the adhesive supply **140** via an adhesive supply line **142** and a lid **120** mounted to the container **110** so as to substantially cover the adhesive reservoir. The adhesive supply **140** includes an opening and a removable cap **141** configured to seal the opening and be in fluid communication with the adhesive supply line **142** such that adhesive within the adhesive supply **140** is able to flow by gravity through the cap **141** and into the supply line **142**.

(40) The adhesive applicator **100** further includes an adhesive roller **130** rotatably coupled to the container **110** and configured to be at least partially submerged in the adhesive reservoir so that upon rotation, adhesive from the reservoir is entrained up and around at least a portion of the adhesive roller **130**. In use, adhesive is transferred from the adhesive roller onto a bottom surface of a block as the block is moved across the roller.

(41) In accordance with the above-described arrangement, adhesive is applied to blocks in a bottom-up manner as opposed to a top-down pumping arrangement in which nozzles dispense adhesive onto the top surface of the block. In a top-down pumping arrangement, before a block can be laid, it must be flipped over so that the surface with adhesive is facing down. During this flipping motion, the adhesive is often prone to dripping which may lead to wastage, inadequate coverage on the block and contamination of the building site. The above-described adhesive application system **10** alleviates this problem and tends to minimise dripping as the block is not required to be rotated after adhesive application.

(42) A roller-based adhesive application system eliminates many parts of a top-down pumping arrangement including pumps, nozzles, nozzle carriages, drives, gears etc. This reduces design complexity and cost. Furthermore, applying adhesive with a roller provides a more consistent and repeatable adhesive signature as less variables (such as line pressure, pump speed etc.) need to be controlled compared to a nozzle-based applicator. A pump-based system can have difficulties dispensing adhesive in a metered dose due to varying viscosity with temperature that can make pumping more difficult. The roller-based system overcomes this issue since adhesive is simply transferred from the roller onto the block as it moves across the roller.

(43) In addition, a roller-based system allows adhesive to be applied easily onto blocks that may have a complex grid or core structure (such as PoroTherm clay blocks) as the roller simply transfers adhesive onto adjacent web structures as the block is moved across the roller, whereas a nozzle application would lead to adhesive being dispensed into cores making it more difficult to achieve consistent coverage.

(44) A number of further examples will now be described.

(45) There are different ways in which adhesive may flow from the adhesive supply to the container of the adhesive applicator via the supply line in order to maintain a desired level in the container for continuous application. In one example, adhesive is provided to the adhesive applicator from the adhesive supply via a gravity feed.

(46) In this example, the adhesive supply is a canister having an opening to allow the canister to be at least partially filled with adhesive; and, a removable cap that seals the opening and is configured to be in fluid communication with the adhesive supply line; wherein, in use, the canister is oriented so that the cap is facing downward and adhesive is able to flow by gravity through the cap and into the supply line. The term canister refers to any suitable type of container, drum or storage vessel that can store the construction adhesive. Typically, it is made from an industrial strength, heavy duty, durable high density polyethylene (HDPE) or similar plastic.

(47) The adhesive supply could take numerous forms including also an adhesive cartridge as described in co-pending application PCT/AU2019/000103 (hereby incorporated by reference) which comprises a bladder for the adhesive, a shell for the bladder, the shell comprising an aperture therethrough, and at least one valve assembly for controlling adhesive flow, the valve assembly comprising a valve body, at least a portion of which extends through the aperture and into the shell.

(48) In an alternative arrangement such as one using the adhesive cartridge as described in co-pending application PCT/AU2019/000103, adhesive from the adhesive supply is pumped into the container via a peristaltic pump. The pump may be servo controlled. Alternatively, in a simpler arrangement a linear peristaltic pump may be used comprising multiple pinch valves which regulate the amount of adhesive pumped into the container.

(49) In respect of the gravity feed system, typically the adhesive supply line includes a valve which is openable to allow flow of adhesive into the container of the adhesive applicator. The valve may be actuated by a motor and controllable to selectively open in order to meter the amount of adhesive supplied to the container in order to maintain a desired level.

(50) Depending on the valve used and torque required to reliably open and close it, in some examples the motor is coupled to a reduction gearbox to increase torque for actuating the valve.

(51) In order to prevent overfilling of the container and also to ensure a sufficient amount of adhesive is in the reservoir for application, the amount of adhesive in the container is constantly monitored. In one example, the amount of adhesive in the container is monitored by at least one of: a level sensor that provides a signal to a controller indicative of the level of adhesive in the container; and, a mass sensor that provides a signal to a controller indicative of the mass of the container.

(52) The level sensor may be a capacitive sensor mounted to the side of the container whereas the mass sensor may be a load cell that is mounted beneath the container to measure the mass thereof.

(53) In another example, a non-contact ultrasonic sensor to measure distance to the top of the adhesive in the container may be used.

(54) In order to optimise adhesive application in all operating environments, it is preferable to heat the adhesive prior to application. This may be achieved by adding a heating element in the base of the container which assists in controlling the viscosity of the fluid.

(55) The adhesive roller itself typically includes an elongate shaft having axles rotatably coupled to opposing ends of the container, and, one or more roller elements coupled to or integral with the shaft. In use, it is the one or more roller elements are at least partially submerged in the adhesive reservoir. The roller elements may be in the shape of circular discs or wheels and preferably with at least one circumferential groove formed therein which assists in entraining a bead of adhesive onto the or each roller element. The roller elements may vary in number and spacing to accommodate desired adhesive signature patterns for different block types and geometries.

(56) In order to transfer adhesive onto the block, the one or more roller elements at least partially protrude above the lid through openings therein.

(57) In one example, the adhesive roller is driven by a motor coupled to the elongate shaft via a coupling or transmission element. In its simplest form, the coupling element may just be a short length of polyvinyl chloride (PVC) tubing that sheaths over the end of the motor shaft at one end and over the shaft of the roller at the other. Driving the roller in this manner is advantageous as it provides a form of self-priming to the system ensuring that adhesive is always entrained onto the

roller whenever a block moves across it. This ensures consistent application along the entire length of the block.

(58) Typically, a bead of adhesive is entrained onto the one or more roller elements as the shaft rotates and is transferred onto the block as the block moves across the adhesive roller. Whilst in some implementations the block physically contacts the one or more roller elements as it moves across, this is not necessary and a gap between the block and the one or more roller elements may be provided that is less than or equal to the thickness of the bead of adhesive entrained on the roller elements.

(59) In one example, the system is mounted to a block laying and adhesive applying head of the automated block laying machine, the head located at a distal end of a telescopingly extendable and foldable boom for conveying blocks therealong. The adhesive application may be adjustably mounted to the head via a support post and bracket to permit the applicator to be height adjusted to suit different block geometries and also in some embodiments adjusted laterally or longitudinally.

(60) In another broad form, an aspect of the present invention seeks to provide a block laying and adhesive applying head for an automated block laying machine, the head mountable to a distal end of a telescopingly extendable and foldable boom for conveying blocks therealong, the head including an adhesive application system including: an adhesive supply; and, an adhesive applicator including: a container for storing an adhesive reservoir received from the adhesive supply via an adhesive supply line; a lid mounted to the container so as to substantially cover the adhesive reservoir; and, an adhesive roller rotatably coupled to the container and configured to be at least partially submerged in the adhesive reservoir so that upon rotation adhesive from the reservoir is entrained up and around at least a portion of the adhesive roller, wherein, in use, adhesive is transferred from the adhesive roller onto a bottom surface of a block as the block is moved across the roller.

(61) In one example, the head includes a block transfer apparatus as described in applicant's co-pending application PCT/AU2021/050361 for transferring a block from a distal end of the boom to a robotic arm having an end effector which grips and lays the block at a predetermined spatial position and orientation, the block transfer apparatus configured to clamp the block and linearly extend it across the adhesive roller before it is transferred to the robotic arm and laid.

(62) In another broad form, an aspect of the present invention seeks to provide a block laying and adhesive applying head for an automated block laying machine, the head mountable to a distal end of a telescopingly extendable and foldable boom for conveying blocks therealong, the head including: a support mountable to the boom; a robotic arm having an end effector operable to grip and lay blocks at a predetermined spatial position and orientation; a block transfer apparatus for transferring a block from the distal end of the boom to the robotic arm; and, an adhesive application system mounted to the support including: an adhesive supply; and, an adhesive applicator including: a container for storing an adhesive reservoir received from the adhesive supply via an adhesive supply line; a lid mounted to the container so as to substantially cover the adhesive reservoir; and, an adhesive roller rotatably coupled to the container and configured to be at least partially submerged in the adhesive reservoir so that upon rotation adhesive from the reservoir is entrained up and around at least a portion of the adhesive roller, wherein, in use, adhesive is transferred from the adhesive roller onto a bottom surface of a block as the block transfer apparatus linearly extends a block across the roller.

(63) In yet a further broad form, an aspect of the present invention seeks to provide an automated block laying machine, including: a base; a telescopingly extendable and foldable boom rotatably mounted from the base for conveying blocks therealong; and, a block laying and adhesive applying head, the head mountable to a distal end of the boom, the head including: a support mountable to the boom; a robotic arm having an end effector operable to grip and lay blocks at a predetermined spatial position and orientation; a block transfer apparatus for transferring a block from the distal end of the boom to the robotic arm; and, an adhesive application system mounted to the support

including: an adhesive supply; and, an adhesive applicator including: a container for storing an adhesive reservoir received from the adhesive supply via an adhesive supply line; a lid mounted to the container so as to substantially cover the adhesive reservoir; and, an adhesive roller rotatably coupled to the container and configured to be at least partially submerged in the adhesive reservoir so that upon rotation adhesive from the reservoir is entrained up and around at least a portion of the adhesive roller, wherein, in use, adhesive is transferred from the adhesive roller onto a bottom surface of a block as the block transfer apparatus linearly extends a block across the roller.

(64) The adhesive application system **10** shall now be described in further detail.

(65) As shown in FIGS. **4A** to **4B**, the adhesive applicator **100** comprises a rectangular container **110** with an aperture **112** in a side thereof which provides an inlet for a fitting of the adhesive supply line **142** in order for adhesive to be metered into the container **110**. The container **110** has slots **111**, **113** in the ends thereof which receive ends **133**, **134** of the elongate shaft **132** of the adhesive roller. In this example a pair of roller elements **130** in the form of circular discs with a slotted circumference are coupled to the shaft **132**.

(66) A lid **120** presses onto the top of the container **110**, the lid **120** having a pair of spaced apart slots **121**, **122** through which the adhesive rollers **130** protrude in operation. A pair of elastic bands or straps **114**, **115** may assist in securing the lid **120** to the container **110** as shown. The lid may have flared sides which overhang the container to assist in directing any adhesive spill into catchment areas. For example, as shown in FIG. **2A**, a disposable drip tray **170** may be located in front of the adhesive applicator to catch any overflow and drips from the bottom of a block as it is being moved across the applicator.

(67) The container **110** is mounted onto a heating plate **102** which is screwed underneath the container for controlling the temperature (and viscosity) of the adhesive in the container before it is applied.

(68) Referring now to FIGS. **1C**, **1E** and **1F**, further detail of actuation of the adhesive roller **130** and supply line valve **144** shall be described. In FIG. **1E**, there is shown a sectional view through the adhesive roller shaft **132**. Servo motor **160** drives this shaft **132** via a coupling element **162** which connects the motor shaft **161** and axle **135** of the roller shaft **132**. In this way, the roller **130** can be primed and always ready to transfer a reliable bead of adhesive onto a block. The rotational speed of the roller **130** can be varied in accordance with a datafile which defines this parameter in order to ensure a reliable and repeatable application.

(69) In an alternative example such as shown in FIG. **16**, a system **700** may be configured so that the block **1** may move across a roller or gear **734** which turns connecting gear **732** and then primes the adhesive roller **730** so that when the block **1** passes over the applicator, a bead of adhesive is already entrained onto the roller **730**. In this example, the block itself primes the system and no motor is necessary (i.e. the adhesive roller does not need to be motor driven).

(70) Adhesive supply line valve **144** connects a first supply line segment **142.1** (between the supply **140** and valve **144**) with a second supply line segment **142.2** (between the valve **144** and container **110**). Adhesive from the supply **140** flows into first supply line segment **142.1** by gravity and is then metered into the second supply line segment **142.2** which flows into container **110** when the valve **144** is open. The valve **144** is pressed into a fitting **154** which has a complementary shape to the exterior of the valve body. The fitting **154** is actuated by a servo motor **150** via a reduction gearbox **152** which increases the torque output of the motor **150** so as to be able to rotate the fitting **154** and thereby open and close the valve **144**.

(71) In operation, a capacitive sensor mounted to the container **110** senses a level of adhesive therein and the valve **144** is then controlled in order to meter the appropriate amount of adhesive into the container **110**.

(72) Additional features which the adhesive application system **10** may have include a metal bump plate **180** installed to the rear of the adhesive applicator **100** slightly lower than the rollers **130** so that if the block is transferred lower than expected, it will contact the bump plate **180** instead of

crashing into the applicator and causing damage to the system. The system may further include an outer enclosure **190** which provides a cover for part of the system and provides further protection to components especially to prevent any unwanted adhesive spill onto other parts of the system.

(73) The adhesive application system **10** is installed onto a block laying and adhesive applying head **3** of an automated block laying robot as depicted in FIGS. **3A** and **3B**. The head **3** is mountable to a distal end of a telescopingly extendable and foldable boom **2** for conveying blocks therealong. A block transfer apparatus clamps a block received from a shuttle in the boom and extends the block across the adhesive applicator and then a laying arm picks up the block and places it at a desired position. The adhesive supply **140** is removably mounted to the side of the head **3** and the adhesive applicator assembly is mounted from the head by a support post.

(74) Another example of an adhesive application system **10'** is shown in FIGS. **5A** and **5B**.

(75) In this example, the system **10'** includes an adhesive supply (not shown) and an adhesive applicator **200** including a container **210** for storing an adhesive reservoir **201** received from the adhesive supply via an adhesive supply line **20**, a pair of sprung roller arms **222** coupled to the container **210** for supporting an adhesive roller **230**. The adhesive roller **230** is rotatably coupled to the sprung roller arms **222** and is configured to extend across the container **210**. The adhesive roller **230** is at least partially submerged in the adhesive reservoir **201** so that upon rotation, adhesive from the reservoir **201** is entrained up and around the roller **230**. In use, adhesive is transferred from the adhesive roller **230** onto a bottom surface of a block **1** as the block **1** is moved across the roller **230**.

(76) The adhesive supply could take numerous forms as previously described such as the adhesive cartridge described in co-pending application PCT/AU2019/000103. Typically, adhesive from the adhesive supply is pumped into the container **210** via a peristaltic pump **22** and supply line **20**. The pump **22** may be actuated by a servo motor **24** as shown in FIG. **5A**. Alternatively, in a simpler arrangement a linear peristaltic pump may be used comprising multiple pinch valves which regulate the amount of adhesive pumped into the container.

(77) The term “sprung roller arms” shall be understood to refer to flexible or compliant arms which support the adhesive roller and which are adapted to bend or flex under load about a pivot or hinge so as to induce positive contact between the roller and block in use. The arms could be plastic or metal leaf spring arms that are integrally moulded or formed with a lid or container cover, or alternatively formed part of components which engage onto the container but may not necessarily cover the container. The arms may be separate components, an integral component or otherwise connected together.

(78) In this example, the pair of sprung roller arms **222** form part of a lid or cover **220** that at least partly conceals the adhesive reservoir **201** in the container **210**. In this regard, it is generally desirable to cover the adhesive so as to prevent it from curing when exposed to air or humidity. In this arrangement, the sprung roller arms **222** may be integrally moulded or formed with the lid or otherwise attached to the lid structure.

(79) The pair of sprung roller arms **222** are disposed about opposing sides of the container **210**. In this manner, the adhesive roller **230** is typically supported proximate opposing ends which allows the roller **230** to rotate. As shown in FIGS. **5B** and **9A**, the pair of sprung roller arms **222** include a groove or slot **222A** into which an axle **231** of the adhesive roller **230** is rotatably coupled.

(80) The components of the adhesive applicator **200** could be made from plastic or metal and in some examples, the container, lid, roller and sprung roller arms may be 3D printed and disposable after use. In one example, the container, lid and/or sprung roller arms are 3D printed plastic components whilst the adhesive roller is metal (i.e. steel).

(81) The adhesive applicator **200** may be adjustably mounted to a support post **30** and bracket or plate assembly **40** which has a series of slots which enables the bracket (and adhesive applicator) to be moved up and down and back and forth as desired. This could be a manual adjustment or via suitable actuators which may be controlled to adjust the relative positioning of the applicator. The

adhesive applicator **200** may be mounted onto a tubular support or bar **260** which extends laterally away from bracket **40**.

(82) In this regard, the container **210** is typically mounted in a cradle **240** and removably secured thereto. The cradle **240** is mounted to the tube or bar **260**. In one example, a heated plate **250** is mounted to the base of the cradle **240** and the container **210** is mounted onto the heated plate **250** so as to control the temperature (and viscosity) of the adhesive in the reservoir **201**.

(83) To maintain adhesive levels in the container **210** (to approximately ± 2 mm) a load cell **262** is used as best shown in FIG. **10B**. The load cell **262** sits beneath the container **210** and provide signal indicative of the mass of the container to a control system which then pumps adhesive into the container to maintain a mass of adhesive (and hence level) between programmable maximum and minimum levels. For the roller to work effectively, it needs to be at least partially submerged in the adhesive. As shown in FIG. **10B**, the load cell **262** is positioned beneath the cradle **240** and used to monitor the weight of the adhesive in the reservoir **201** so as to maintain a programmed level of adhesive in the reservoir **201**. Another option utilises a non-contact ultrasonic sensor to measure distance to the top of the adhesive or a capacitive sensor to determine a level of adhesive in the container.

(84) Typically, one or both of the sprung roller arms **222** are coupled to the container **210** by an integral moulding that includes a built-in hinge or pivot **221** (see FIG. **6C**) that allows the arms to flex or bend. In this regard, the arms act like flexibly resilient cantilever members. The moulding (which may be the lid **220**) typically includes a peripheral slot that allows the moulding to be engaged onto a portion of an upper wall of the container. The built-in hinge or pivot **221** may be defined by a slot in the moulding having a terminal end proximate the base of each arm **222** about which the arm **222** can flex.

(85) The adhesive roller may be provided in numerous forms. In the example shown in FIGS. **5A** and **5B**, the adhesive roller **230** has an elongate body having a plurality of circumferential grooves **232** spaced apart along the length of the body. Instead of grooves, the roller surface may alternatively be knurled or have some other kind of surface roughness or pattern applied to it which assists in both preventing roller slippage and providing a desired adhesive signature onto the block. In this example, a single roller is provided which extends substantially across the container **210**.

(86) Referring to FIGS. **6A** to **6C** and FIGS. **7** and **8**, an example of a lid **220** is shown in more detail in which a scraper **270** is coupled to the pair of sprung roller arms **222** proximate a distal end thereof (proximate edge **226** of container **210**), the scraper **270** having an elongate body **271** defining one or more ridged sections **272** including a plurality of ridges **273** operable to block particular grooves of the adhesive roller **230** and scrape adhesive therefrom. Additionally, one or more voids **274** are positioned between the ridged sections **272**, the voids **274** providing areas where adhesive is able to be entrained onto the roller **230** to thereby define a pre-defined adhesive signature. The scraper **270** may be removable and adapted to suit the particular adhesive signature required by selectively allowing adhesive to be entrained onto certain areas of the roller **230**. In one example, the scraper **270** is press fit between the distal ends **222C** of the sprung roller arms **222**. In another example, the lid may act as a mask around the leading edge of the roller, the distance of the edge of the mask away from the roller proportional to the amount of adhesive which is entrained onto the roller. The mask can therefore control the volume of adhesive picked up by the roller and assist in minimising the surface area of adhesive exposed to the environment.

(87) Referring now to FIGS. **9A** to **9D**, it may be desirable to pre-load or tension the sprung roller arms **222**. Once the scraper **270** has been assembled onto the roller arms **222**, pre-tensioning may be achieved by manipulating the arms **222** and urging the scraper **270** beneath tabs **227** located along a lengthwise edge of the lid **220** (see FIG. **9D**). This provides a pre-load when the roller is in contact with the brick to eliminate roller slippage.

(88) The lid **220** is coupled to the container **210** in order to substantially conceal the reservoir **201**. The lid **220** may include a pivotable cover **224** disposed between the pair of sprung roller arms **222**

and extending towards the adhesive roller **230**. The cover **224** is adapted to hinge about built-in pivot point **223**. A distal end **224A** of the pivotable cover **224** may be urged beneath tabs **222B** disposed on the sprung roller arms **222** as shown in FIG. **9C** so as to allow any unused adhesive on the surface of the cover **224** to drain back into the reservoir **201**.

(89) FIGS. **11A** to **11H** show FIGS. **11A** to **11H** provide a sequence of views illustrating a block **1** moving across the roller **230** of the adhesive applicator **200** to thereby apply adhesive to the bottom surface **2** of the block **1**. In this example, the bottom surface **2** of the block **1** contacts the roller **230** in order to transfer the adhesive entrained on the roller onto the surface of the block. However, in alternative examples, since a laminar bead of adhesive having a thickness is entrained onto the roller, it is not necessary for there to be physical contact between the block and roller and a gap may be provided less than or equal to the thickness of the bead.

(90) As the sprung roller arms **230** are compliant and able to hinge about pivot **221**, as the block moves across the top of the roller, the roller may move up and down in a compliant fashion to conform to the block shape which may vary along its length.

(91) Examples of block adhesive signatures are shown in FIGS. **12A** to **12C**. In FIG. **12A**, a concrete masonry unit (CMU) block **300** is shown of cuboid form having sides **301**, **303** and ends **304**, **305** and a plurality of cores **306** spaced lengthwise along the block. Two parallel beads of adhesive **301**, **312** are applied lengthwise along the bottom surface **302** of the block adjacent sides **301**, **303**. In FIG. **12B**, a slightly wider CMU block **400** is shown having sides **401**, **403** and ends **404**, **405** and a plurality of cores **406** spaced lengthwise along the block. Two parallel beads of adhesive **410**, **420** are applied lengthwise along the bottom surface **402** of the block adjacent sides **401**, **403**. Adhesive is also applied to the ribs between the cores **406** as depicted by beads **430**, **440**. These additional beads provide greater adhesion across the width of the block for a wider block such as this. Finally, in FIG. **12C** a clay block **500** is shown having a complex core arrangement. Block **500** is shown having sides **501**, **503** and ends **504**, **505** and a plurality of cores **506**. Four parallel beads of adhesive **510**, **520**, **530** and **540** are applied lengthwise along the bottom surface **502** of the block along longitudinal ribs or stiffeners. These different adhesive signatures are formed by rollers having roller elements spaced apart at different intervals with variable roller element widths dependent on block type.

(92) In this regard, the adhesive roller may include an elongate shaft and a plurality of spaced apart roller elements coupled to the shaft. The roller elements are typically cylindrical and include a plurality of circumferential grooves. Optionally, the roller elements may be spaced apart by spacer elements and configured as appropriate in order to obtain the desired adhesive signature for a particular block type.

(93) In yet a further example, the adhesive roller includes alternating roller and spacer elements coupled to the shaft and wherein the spacer elements have a diameter less than the roller elements. In this example, the roller and spacer elements would be thin discs that are arranged so that the spacers act like grooves. In accordance with the above examples, the adhesive roller may be reconfigured using a standard set of roller and/or spacer elements as appropriate in order to achieve a desired adhesive signature.

(94) Examples of alternative roller configurations are provided in FIGS. **17A** to **17F**. In FIG. **17A**, there is shown an elongate cylindrical roller having a knurled body **810**. In FIG. **17B**, there is shown a roller **900** having an elongate shaft **901** and a pair of spaced apart roller elements **910** each with a plurality of spaced circumferential grooves **911**. In FIG. **17C**, there is shown a roller **1000** having an elongate shaft **1001**, and a plurality of spaced apart roller elements **1010** that are separated by spacer tubes **1012**. In FIG. **17D**, there is shown a roller **1100** having an elongate shaft **1101** and a plurality of thin roller elements **1110** separated by thin spacers **1112**. FIG. **17D** shows a roller **1200** having an elongate shaft **1201** and a central roller element **1210** which is intended to be masked as shown in FIG. **17E** by lid **1410** which acts as a mask around the leading edge **1412** of the roller, the distance of the edge of the mask away from the roller proportional to the amount of

adhesive which is entrained onto the roller. The mask can therefore control the volume of adhesive picked up by the roller and assist in minimising the surface area of adhesive exposed to the environment.

(95) The above described adhesive system is for use with an automated block laying machine **600** as shown in FIGS. **13A** and **13B** and FIGS. **14** and **15A** to **15C**. The automated block laying machine **600** may include a base **610** (such as a truck or other vehicle). A telescopically extendable and foldable boom **620** for conveying blocks is rotatably coupled to the base and boom section **622** and **624** are coupled about a pivot joint. A block laying and adhesive applying head **630** is mounted to the distal end of the boom **620** and having a block laying robot arm **632** and end effector **633** for placing a block in a desired position. A block transfer apparatus **636** clamps a block **1** received from the distal end of the boom **620** and linearly extends it across the adhesive applicator **200** mounted to the head **630** as shown in FIGS. **14** and **15A** to **15C**. The adhesive supply **50** is shown mounted to the side **635** of the head **630**. In this way, adhesive is applied to the bottom surface of the block **1** before the laying robot **632** picks up the block and lays it.

(96) Throughout this specification and claims which follow, unless the context requires otherwise, the word “comprise”, and variations such as “comprises” or “comprising”, will be understood to imply the inclusion of a stated integer or group of integers or steps but not the exclusion of any other integer or group of integers. As used herein and unless otherwise stated, the term “approximately” means $\pm 20\%$.

(97) Persons skilled in the art will appreciate that numerous variations and modifications will become apparent. All such variations and modifications which become apparent to persons skilled in the art, should be considered to fall within the spirit and scope that the invention broadly appearing before described.

Claims

1. An adhesive application system for an automated block laying machine, the adhesive application system including: an adhesive supply; an adhesive applicator including: a container for storing an adhesive reservoir received from the adhesive supply; a lid mounted to the container so as to substantially cover the adhesive reservoir; and, an adhesive roller rotatably coupled to the container and configured to be at least partially submerged in the adhesive reservoir so that upon rotation adhesive from the adhesive reservoir is entrained up and around at least a portion of the adhesive roller, and an adhesive supply line connected to and movable with the container, wherein the adhesive supply line is configured to receive adhesive from the adhesive supply and convey the received adhesive to the container, wherein, in use, adhesive is transferred from the adhesive roller onto a bottom surface of a block as the block is moved across the adhesive roller.
2. The adhesive application system according to claim 1, wherein the adhesive supply is configured to provide adhesive to the adhesive applicator via a gravity feed.
3. The adhesive application system according to claim 2, wherein the adhesive supply is a canister having: an opening to allow the canister to be at least partially filled with adhesive; and, a removable cap that seals the opening and is configured to be in fluid communication with the adhesive supply line; wherein, in use, the canister is oriented so that the cap is facing downward and adhesive is able to flow by gravity through the cap and into the supply line.
4. The adhesive application system according to claim 3, wherein the adhesive supply line includes a valve which is openable to allow flow of an amount of adhesive into the container of the adhesive applicator.
5. The adhesive application system according to claim 4, wherein the valve is configured to be actuated by a motor so as to selectively open in order to meter the amount of adhesive supplied to the container.
6. The adhesive application system according to claim 5, wherein the motor is coupled to a

reduction gearbox to increase torque for actuating the valve.

7. The adhesive application system according to claim 4, wherein the adhesive application system is configured to monitor the amount of adhesive in the container via: a level sensor configured to provide a signal to a controller indicative of the level of adhesive in the container; and/or, a mass sensor configured to provide a signal to a controller indicative of the mass of the container.

8. The adhesive application system according to claim 7, wherein: the level sensor is a capacitive sensor mounted to a side of the container; and/or, the mass sensor is a load cell mounted beneath the container.

9. The adhesive application system according to claim 1, wherein: the adhesive application system is configured to heat the adhesive prior to application; and/or, a heating element is provided at a base of the container.

10. The adhesive application system according to claim 1, wherein the adhesive roller includes: an elongate shaft having axles rotatably coupled to opposing ends of the container; and, one or more roller elements coupled to or integral with the shaft.

11. The adhesive application system according to claim 10, wherein: in use, the one or more roller elements are at least partially submerged in the adhesive reservoir; and/or, the one or more roller elements at least partially protrude above the lid through openings therein.

12. The adhesive application system according to claim 10, wherein: the one or more roller elements are circular discs or wheels; and/or, the one or more roller elements have at least one circumferential groove formed therein.

13. The adhesive application system according to claim 10, wherein the adhesive roller is configured to be driven by a motor coupled to the elongate shaft via a coupling element.

14. The adhesive application system according to claim 10, wherein the adhesive roller is configured to entrain a bead of adhesive onto the one or more roller elements as the shaft rotates and to transfer the bead of adhesive onto the block as the block moves across the adhesive roller, and wherein the one or more roller elements of the adhesive roller are configured to contact the block and/or provide a gap between the block and the one or more roller elements that is less than or equal to the thickness of the bead of adhesive.

15. The adhesive application system according to claim 1, wherein the adhesive application system is mounted to a block laying and adhesive applying head of an automated block laying machine, the head located at a distal end of a telescopingly extendable and foldable boom for conveying blocks therealong.

16. The adhesive application system according to claim 15, wherein the adhesive applicator is adjustably mounted to the head via a support post and bracket.

17. A block laying and adhesive applying head for an automated block laying machine, the head mountable to a distal end of a telescopingly extendable and foldable boom for conveying blocks therealong, the head including an adhesive application system including: an adhesive supply; and; an adhesive applicator including: a container for storing an adhesive reservoir received from the adhesive supply; a lid mounted to the container so as to substantially cover the adhesive reservoir; and, an adhesive roller rotatably coupled to the container and configured to be at least partially submerged in the adhesive reservoir so that upon rotation adhesive from the adhesive reservoir is entrained up and around at least a portion of the adhesive roller, and an adhesive supply line connected to and movable with the container, wherein the adhesive supply line is configured to receive adhesive from the adhesive supply and convey the received adhesive to the container, wherein, in use, adhesive is transferred from the adhesive roller onto a bottom surface of a block as the block is moved across the adhesive roller.

18. The block laying and adhesive applying head according to claim 17, wherein the head includes a block transfer apparatus for transferring a block from a distal end of the boom to a robotic arm having an end effector which grips and lays the block at a predetermined spatial position and orientation, the block transfer apparatus configured to clamp the block and linearly extend the

block across the adhesive roller before the block is transferred to the robotic arm and laid.

19. The block laying and adhesive applying head according to claim 17, the head further including: a support mountable to the boom, wherein the adhesive application system is mounted to the support; a robotic arm having an end effector operable to grip and lay blocks at a predetermined spatial position and orientation; and, a block transfer apparatus for transferring a block from the distal end of the boom to the robotic arm, wherein, in use, adhesive is transferred from the adhesive roller onto the bottom surface of the block as the block transfer apparatus linearly extends the block across the roller.

20. An automated block laying machine, including: a base; a telescopingly extendable and foldable boom rotatably mounted from the base for conveying blocks therealong; and, a block laying and adhesive applying head, the head mountable to a distal end of the boom, the head including: a support mountable to the boom; a robotic arm having an end effector operable to grip and lay blocks at a predetermined spatial position and orientation; a block transfer apparatus for transferring a block from the distal end of the boom to the robotic arm; and, an adhesive application system mounted to the support including: an adhesive supply; an adhesive applicator including: a container for storing an adhesive reservoir received from the adhesive supply; a lid mounted to the container so as to substantially cover the adhesive reservoir; and, an adhesive roller rotatably coupled to the container and configured to be at least partially submerged in the adhesive reservoir so that upon rotation adhesive from the adhesive reservoir is entrained up and around at least a portion of the adhesive roller; and an adhesive supply line connected to and movable with the container wherein the adhesive supply line is configured to receive adhesive from the adhesive supply and convey the received adhesive to the container, wherein, in use, adhesive is transferred from the adhesive roller onto a bottom surface of a block as the block transfer apparatus linearly extends a block across the adhesive roller.

21. An adhesive application system for an automated block laying machine, the adhesive application system including: an adhesive supply; and; an adhesive applicator including: a container for storing an adhesive reservoir received from the adhesive supply; a lid mounted to the container so as to substantially cover the adhesive reservoir, the lid having one or more openings therein; and, an adhesive roller rotatably coupled to the container and including one or more roller elements configured to be at least partially submerged in the adhesive reservoir so that upon rotation adhesive from the adhesive reservoir is entrained up and around at least a portion of the one or more roller elements, and an adhesive supply line connected to and movable with the container, wherein the adhesive supply line is configured to receive adhesive from the adhesive supply and convey the received adhesive to the container, wherein the one or more roller elements at least partially protrude above the lid through the one or more openings of the lid so that, in use, adhesive is transferred from the adhesive roller onto a bottom surface of a block as the block is moved across the lid and the one or more roller elements.
